LEARNING DATA VISUALISATION SUBJECT USING AUGMENTED REALITY

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Thesis submitted in fulfilment of the requirements for Bachelor of Computer Science (Hons.) Multimedia Computing
Faculty of Computer and Mathematical Sciences

January 2020

SUPERVISOR APPROVAL

LEARNING DATA VISUALIZATION SUBJECT USING AUGMENTED REALITY

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06 JULY, 2020

STUDENT DECLARATION

I certify that this thesis and project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise are fully acknowledged in accordance with the standard referring practices of the discipline.

……..

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ACKNOWLEDGEMENT

In The Name of Allah, The Most Beneficent, The Most Merciful.

I would like to express my sincere gratitude to several individuals for supporting me throughout my graduate study. First, I wish to express my sincere gratitude to my supervisor, Dr Zainura Idrus, for her enthusiasm, patience and insightful comments, practical advice and unceasing ideas that have helped me tremendously at all times in my research and writing of this thesis. Without her support and guidance, this project would not have been possible.

I also wish to express my sincere thanks to my final year project lecturer, Dr Marina Ismail Whom never get tired of being a teacher, guidance throughout the lessons. Special thanks to my beloved parent and family, whom always give full support and motivation in becoming a better person. Thank you for all the encouragement support me until the project is completed.

ABSTRACT

Data visualization is a process of converting big data into visualized form. It requires analysis approach to extract hidden knowledge. The main purposes are to assist in decision making, planning and prediction processes. Mostly, students in FSKM learning this data visualisation subject in a text based form and students tends to find difficulties in understand this subject. Therefore, there is an emerging method in making learning more interesting and more deliverable which is Augmented Reality. Augmented Reality is a new technology that brings virtual environment to reality environment. This is a new way of learning to approach student interested and gain understanding towards this subject. To gain more students understanding, addition features such taking quizzes after learning session will develop their understanding skills towards the subject.

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CHAPTER 1

INTRODUCTION

This chapter explained the rationality and background of the study. Also, the problems in this study are discussed and objectives are drawn based on the problems. Other than that, there is the details and significance of this study research are also have been listed.

1.1 BACKGROUND OF STUDY

Education is generally known as transferring knowledge from one person to another. The process of transferring knowledge can be in many ways. One of the learning processes is applied in technology education. The reason for integrating technology into learning process is to upgrade student learning outcomes, approach the learning's access and to boost learner motivation (Bower,2017). Moreover, this way is more cognitive rather than static learning because it is more real as learner can understand and visualize it better.

Nowadays, learning can be in two ways. Is either face to face class or non-face to face. But now more and more education is going towards non-face to face. Such as, the online learning. The use of online learning is incorporate into governmental plan and programs that persuade technologies in education (Tenorio, BittenCourt, Isotani & Silva,2016). Students can access at any time and make easy to access whenever they are available through online learning (Aldholay, Issac, Abdullah,
Abdulsalam & Al-Shibami,2018). This will help student to learn in their own pace.

Hence, there is a lot of technique used for learning process. For instance, some use videos, some apply multimedia element and some use AR in teaching. This method usually uploads on web based. However, AR is much better than video and multimedia technique is because AR is more cognitive. Currently, in Faculty of Science Computer and Mathematics (FSKM) visualization subject teaching in a static way which only used power point method. Video method used to project the method. There is a lack in showing the way of teaching in 3D manner.

However, there is an emerging method for learning that is Augmented Reality (AR). Augmented Reality is a technology that can brings reality into virtually environment. (Hady, Francisco,2019). The process of learning and gaining information from a topic can be more interesting in education field (Hady, Francisco,2019). At this point, most of the teaching in FSKM are using blended learning which there is face to face class and non-face to face class. Although, non-face to face is more towards the forum, online test, quiz and assignment but there is still lacking of online teaching in the form of AR in UiTM.

1.2 PROBLEM STATEMENT

The usage of Augmented Reality can increase students learning motivation and enhance their academic achievement (Tasneem,2019). Furthermore, AR technologies brings meaningful learning for student (Tasneem,2019).

The usage of AR has been applied to some universities for educational purposes. For instances, students can learn character in the story that are appeared virtually in historical subject (Hady,2019). Other than that, learning train technology which fascinated student to study by using augmented reality application where the train is represented in 3D virtually (Hady,2019). As for now, UiTM mission is to enhance the knowledge and expertise students in all fields of study. However, UiTM is still lacking of this kind of teaching mode. Eventhor, UiTM still have the online teaching but there is still lacking of teaching. Thus, this project is going to develop Augmented Reality prototype in a virtual education for teaching one of the subjects which is DSC651.

1.3 OBJECTIVE

In order to apply interactive way of learning in Augmented Reality, three objective must be achieved which are:

- To analyze various Augmented Reality technique in the area of education.
- To develop an education prototype in Augmented Reality.
- To test the functionality of the prototype.

1.4 PROJECT SCOPE

This project will be focusing on one chapter from DSC651, Data visualization subject which is called as basic visualization design.

1.5 Significance

The result from this project will allow the student to study in their own without limitation of time and elsewhere. Moreover, student can learn at their own pace as student can repeat the subject again and have control to the system. This kind of system will help student to visualize what they learnt much better.

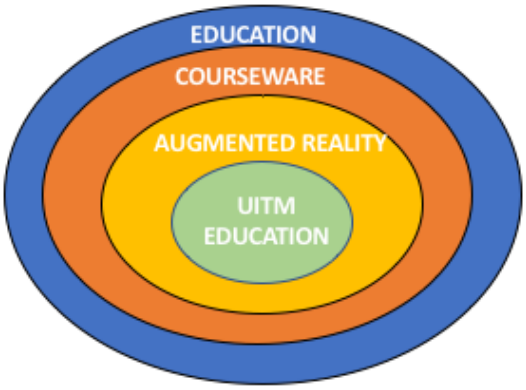
1.6 Summary

As a conclusion in this chapter, background studies of this project are briefly explained. Then, important of this project also have been highlighted to see an overall view of the problems occur for the project and find the right solution to overcome it. Other than that, there is few objectives shown about what to achieve for this project. Then, the scope and significance of the project are also have been discussed in this chapter.

LITERATURE REVIEW

CHAPTER 2

In this chapter, an overview of the important elements from past studies will be shown and briefly described. Most of these studies have been selected from few resources such as journal, article and other resources that are related to this project development. This chapter will cover topics on education, courseware, UiTM education system and Augmented Reality,



2.1 EDUCATION

Learning is commonly known as transferring knowledge from one person to another. Learning is about making changes that can brought one person to develop a new skill and be knowledgeable. Nowadays, learning can be acquired through various ways. One of them is through online learning where it become a medium for education field such as blended learning, e-learning and distance learning (Clark & Mayer, 2016). Since it is beneficial, most government in most countries encourage the use of online learning as a strategy for advancement in education of their countries (Isaac, Aldholay, Abdullah, Ramayah, 2019).

Learning can be performed in two ways. It is either face to face in a classroom mode or non-face to face through various online mediums. Today, more and more education institutes are going towards non face to face learning mode. In recent years, many universities offer online courses as a medium of learning, instead of the traditional classes. Online learning and online schools have transformed the educational system. (Stan, 2014)

Through online learning, students are more flexible and easy to access notes at anytime and from anywhere (Aldholay, Isaac, Abdullah, Abdulsalam & Al- Shibami , 2018). This is also a new opportunity for students to interact and communicate easily with a bigger base learners and able to take time to understand any information deeply (Aldholay, Abdullah, Abdulsalam & Al- Shibami, 2018).

Furthermore, online learning make possible for people who are far from cities to go to universities (Muthanna & Karaman, 2014 ; Jung & Lee,2018). Online learning also help students as such students have freedom to learn on their own pace. Therefore, this new technology bring positive impact to education. It brings benefits to students through the latest revolutionary trends (Stan, 2014).

2.2 On-line Learning System in UNIVERSITY TECHNOLOGY MARA (UiTM)

Currently, UiTM has an online learning system which is known as i-Learn. i-Learn is a platform for students and lecturers to communicate for educational purposes. Most of UiTM students used this system to access notes given by lecturers, discussion and assignment submission. Furthermore, there is also a non-face to face class which is refereed to as blended learning. Thus, FSKM being one of faculty in UiTM, currently is using blended learning as one of learning method. Most of students in FSKM use this blended learning platform for various learning activities such as online study and online assessment which involve tests and quizzes. However, some students find difficulties in understanding the notes and easily felt bored by only watching online educational video. This is because the learning method is static and lacking of interactive elements.

Therefore, there is an emerging method in learning which is more interactive and it is called augmented reality(AR).

2.3 AUGMENTED REALITY IN EDUCATION (AR)

Augmented Reality (AR) is a technology that combines real and virtual objects in a real environments which appear in the form of three dimensions. This is where virtual objects are immersed into a real world through time interaction and three-Dimensional (Layona, Yulianto, Tunardi, 2018). AR technology has been growing fast throughout years as this technology is broadly used in many fields (Pranoto, panggabean, 2019). Other than that, AR is a useful support for intelligence. They could assist users in gaining more understanding of problem solving and task performing (Jo, Kim, 2019). In addition, AR could be supported by various platforms, such as computers, tablets and smart phones. These devices capture the real environment and display them virtually. In term

2.4 AR TECHNIQUES

In term of technique, there are a few techniques used to implement this augmented reality. The techniques are Marker-Based, Marker-Less, Projection-based, Super Imposition based and Recognition based AR.

2.4.1 MARKER-BASED

Marker-based is also known as Image Recognition. It uses camera or a visual marker to display an output (Noida,2019). The technique is commonly used for QR code and 2D code. Marker-based captures real-world scene through camera, where markers are added to the scene. Then, the processing unit generates and renders the virtual contents over the captured scene (Dash, Behera, Dogra, Roy, 2018). In addition, any calculation on the objects position or orientation can be done through the markers which is detectable by camera (Noida, 2019).

2.4.2 MARKER-LESS

Marker-less is the most widely implemented AR technique which is also known as Location-based AR or GPS. Currently, marker-less AR provides extra information about an object by providing users with different transformation and orientation (Ziaei, Hahto, Siuko, Semeraro, 2014). Moreover, marker-less AR helps improving tracking accuracy by enabling real-time performance on the environment (Macedo, 2015).

2.4.3 PROJECTION BASED

Projection based works as a detector of an object which located in front of a camera and later displays information about the objects on screen. What it does is capturing real world objects which are then projected through artificial light. The projection generates 3D interactive hologram. In addition, it allows human interaction by the sense of touch (Noida, 2019).

2.4.4 SUPERIMPOSTION

Superimposition in AR is about replacing view of an object in a specific place.
In such situation, the original view of an object is partially or fully replaced with the same object but in an augmented view (Noida, 2019). For example, IKEA – augmented reality furniture catalogue that augments furniture virtually onto real floor in order to make the room more alive (Noida, 2019).

2.4.5 LOCATION BASED

Basically, location based AR in most cases are used for AR location browsers which help users to discover interesting places within their current location. There are some additional characteristics that support this location based and immersion. They are then, applied in mobile and location based interfaces. The combination of physical and digital space, bring interaction into physical world (De Souza E Silva & Delacruz,2006). Not only that, engagement, engrossment and total immersion is one of the context in location based AR which details out that engagement (Georgiou & Kyza, 2017).

2.5 Comparisons of AR technique

This section compares the five Augmented Reality which have been discussed above. Their aims, weaknesses and strengths are also highlighted in the section.

Table 2.1 shows comparison between technique in their aim, strengths and weakness.

Characteristic/ Features	**Marker-based**	**Marker- less**	**Projection- based**	**Superimposition-based**	**Location-based**
Aim	Marker as recognition tool	Provide extra information-based on location	Projecting artificial light onto real-world objects and surface	Replacing view of the object in a specific place.	Discover - places information within their current location
Strengths	High accuracy	No marker needed	Objects can be create for larger deployments	Virtual object replacement anywhere	Contextual information provided
Weakness	Depend only on marker	Lower accuracy	Analyze depth of the object in details.	Requirement of object recognition and limitation.	Depending on suitability of large-scale environment

Based on all techniques compared above, it seems that Marker-Based technique are more suitable to be use because marker-based only requires a marker which is using marker has higher accuracy and much easier for detection compared to other techniques. This marker-based are easier to detect as this technique used marker as recognition tools.

Based on table figure 2.1, a new comparison is derived which are listed in table 2.2 below.

Table 2.2 shows summary of all comparison technique that have been discussed

Technique	**Marker-based**	**Marker-less**	**Projection-Based**	**Super-imposition**	**Super-imposition**
Use Marker	yes	no	no	no	no
Use OR code or Scanner Code	yes	yes	no	no	yes
Able to change position and orientate the object	yes	no	yes	yes	no
Use artificial light	no	no	yes	yes	no
Display information	yes	yes	no	no	yes
Able to change object from one place to another	no	no	yes	yes	no

Domain	Education	Global Positioning System (GPS)	7D-Hologram	IKEA - Augmented Reality Furniture Catalog	Information of specific location such as specific sport location

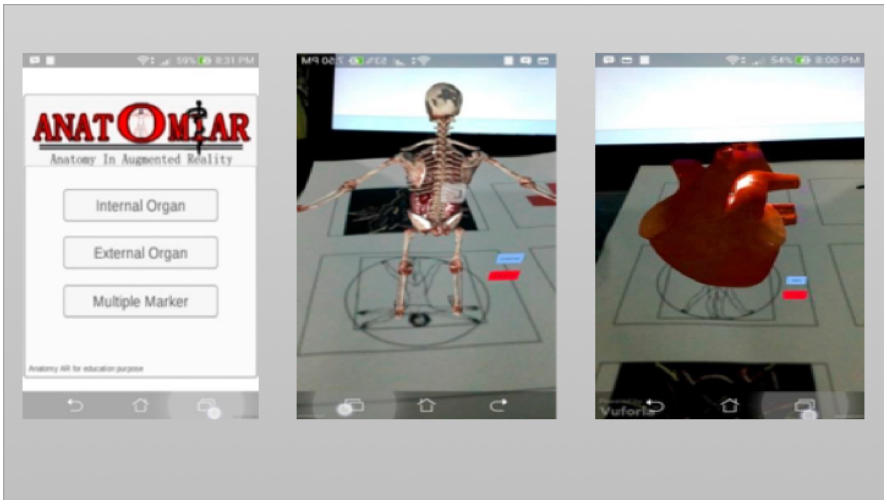
Based on the figure above, each technique has its own specialty towards the application system. Most of this technique have been apply in many areas.

Based from the analysis, the most suitable technique to be used for this project is marker-based technique. This is because marker based is much easier and flexible to be used in e-learning environment. Moreover, marker-based contains a marker which is a detector for camera to capture and recognize the image. With marker, recognition of image is much easier. Also, any position and orientation can be seen immersivity. Not only that, information and description will appear once the camera capture the object as such it helps learners to read and visualize the information well.

2.6 SIMILAR APPLICATIONS

There are also other similar applications that used marker based as a technique in designing e-learning AR systems. Some of the systems are AR Human Anatomy, Fen AR Application and Augmentative Author System. Depicted below is the analysis on the systems.

I. This is a Web based application for medical students to simplify the learning process about human anatomy through Augmented Reality (AR). Medical students are able to visualize human body parts when the body part appeared to the screen. Instead of learning it in theory, medical students are able to understand the content more deeply. It is easier for the students to understand human body part deeply as they can learn from what they see thoroughly.

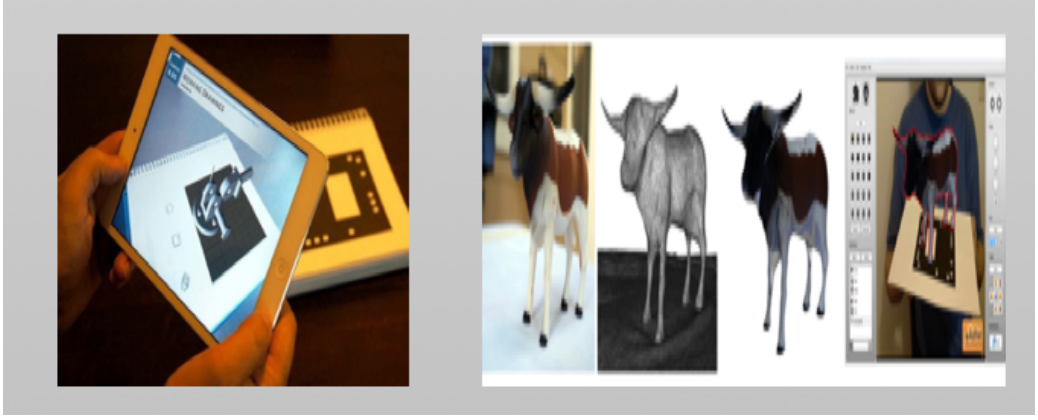


II. **FenAR application** (Fidan & Tuncel, 2019)

FenAR is a software that develop students understanding in learning physics subject. This learning process is based on Marker-based AR and to support Project Based Learning (PBL) in classroom. this application is slightly to help students learn physics subject more attractively.

III. **Augmentaty Author System** (Jorge, manuel,2015)

An educational system to gives experiences in learning engineering design graphics. This system is focus on engineering student. It is to help students to visualize the contents and experience it virtually in order to gain more understanding.



Based on **Table 2.3,** a comparison of each similar application that shown below is based on what each system application has or do not have based on its features.

Table 2.3 Comparison for what each system haves and do not have

Characteristic/ Features	**AR Human Antonomy**	**FEN AR Application**	**Augmentative Author System**	**Basic Visualization Design (DSC651)**
Using "marker"	yes	yes	yes	yes
Able to change the position and orientation of virtual object	yes	yes	no	yes
Three-dimensional object that captured by camera	yes	yes	yes	yes
Clickable of directive buttons	no	no	yes	yes
Zooming and zoom out from different angles	yes	no	yes	yes
Moving from one angles to other angles	yes	no	yes	yes
Calculation of size, distances and orientation of marker	no	yes	no	yes
Domain	Medical Students	Physics Students	Engineering Students	Data Science Students
Platform used	Android smartphone	Tablets and computer	Mobile and desktop based	Mobile/ Smartphone

In **Table 2.3** show that the existing application are lacking some features. Therefore, in Basic Visualization Design (DSC651) system will be included all the features that the other application is lack of.

2.7 SUMMARY

Augmented reality has been widely used and applied in worldwide. Especially in education field. Augmented Reality plays an important role in education system in order to enhance the new way of learning mode for students. Especially students in FSKM UiTM. With this new current technology, students will develop their cognitive thinking skills and be more knowledgeable in visualizing the learning content.

In a prototype development, Marker based will be choose as an augmented reality technique because the technique itself is easy for students to use whereby students will only need the camera to capture the learning material. Marker-Based is one of the attractive ways to be apply in learning as it helps student to understand and visualize more rather than just text based or a theory.

CHAPTER 3

METHODOLOGY

This chapter describes the project methodology as the procedures used in developing the prototype. It consists of the project research framework and workflow that submitted to the development phase.

3.1 Project Research Framework

This chapter describes the project methodology as the procedures used in developing the prototype. It consists of the project research framework and workflow that submitted to the development phase.

In this final year project, Agile model has been applied as the software project research framework. This model is more suitable as it brings more involvement in the development process to the user. According to (Zhang & Patel, 2011), The user collaborative in the agile model could reduce the developer's guess-work. This is because the agile model consists of iterations of the processes. It took 2-3 weeks for user requirements with feedback for each iteration. The changes and incremental can be made as many as the developer sees fit as per user requirement. The nature of the agile model allows users and developers to collaborate in making any additional new function that is necessary for development processes. Other than that, any unnecessary features can also be removed. However, this agile method could bring additional costs to the users but also giving more acknowledgment to the user's opinions.

Nevertheless, the problem with this method is very dependent on the user's understanding of the system to be developed. The effectiveness of the prototype will be affected if the user is uncleared or cannot understand the system. This will situation may lead leads to a misunderstanding of the developer side and also brings to unsatisfied project requirements.

Generally, this agile model is divided into four stages namely Requirement Analysis, Prototype Design and Development, Testing, and Result Analysis. **Figure 3.1** shows the structure of the agile model.

Figure 3.1 Agile Model

Based on figure 3.1 shown above, the details about the figure are discussed in table 3.1

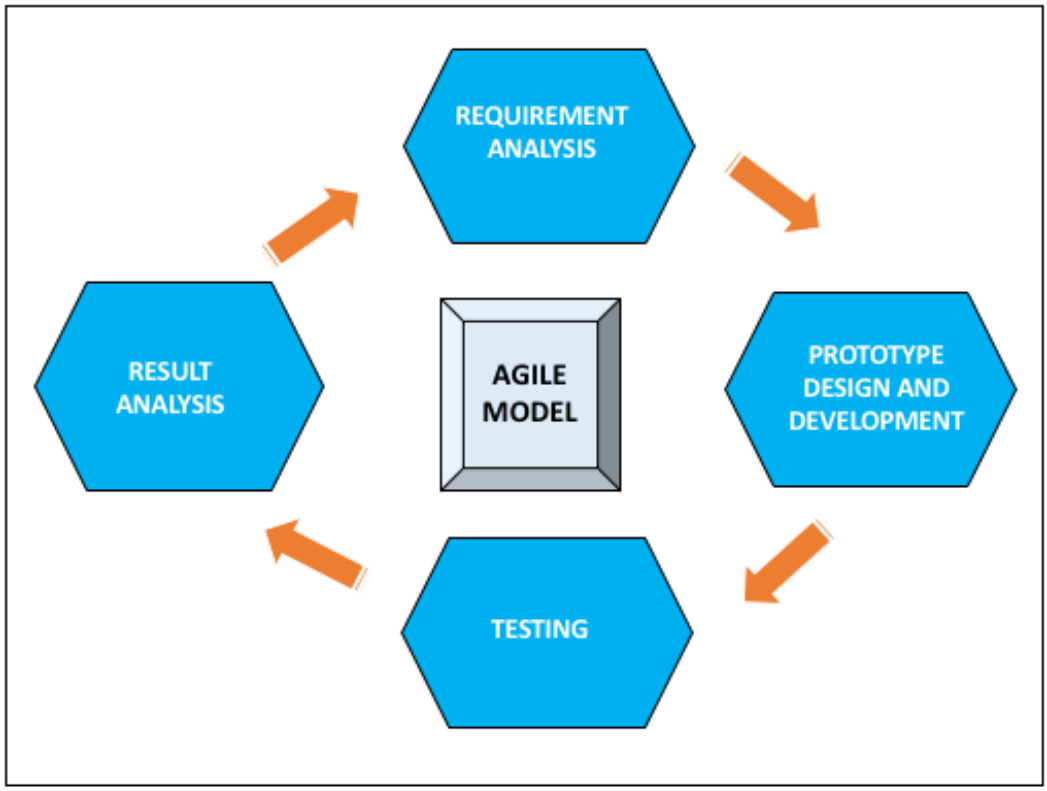


Table 3.1 The details of Agile model

Process	**Description**	**Deliverable**
Requirement Analysis	Project motivation: Developing an augmented reality system for teaching data visualization	• Discussion on the problem statement, aim objective, scope, and significance in Chapter 1.
	• Literature review	• Related journals for Additional references to complete Chapter 2 which all methods have been discussed.
Designing a prototype system and development	• User interface design • Process of each progress of the prototype • Software and hardware requirements	These deliverables are explained more in Chapter 3 • Context diagram and process flow diagram • Executable code module. • Design document. • Codes listing
Testing	• Functionality Testing • User Testing	• The functional system and system delivered meets the user's requirements are explained in Chapter 4.
Result Analysis	Result analysis from testing	• The result from the Testing phase is analyzed

3.2 Requirement analysis

The first task in this phase requires a student as a developer to find and choose a lecturer as his or her supervisor based on their expertise that is related to the project. Then both parties need to sign an undertaking letter, so-called F1, which sort of agreement that applied until the project is completed. The next task is to select the project title. The project requirements are determined through an interview with Dr. Zainura Idrus, lecturer in UiTM Faculty of Computer Science and mathematics.

Next, all information was gathered from the collection of journals, books, and articles that are related to augmented reality, various techniques used, and other resources that are related to the project. Various techniques have been used for augmented reality which is marker-based, marker-less, super-imposition, projection-based, and location-based. Thus, only one technique that has been chosen and the most important process is to discover a marker-based technique that is suitable for the project requirement.

3.3 Designing a prototype system and development

In this phase, the prototype of the mobile application system will be developed. In this project, the function of the system must be well defined. This section gives description on the architecture design of prototype.

3.3.1 Architecture Design

This section is also explained about the architecture design of the prototype.

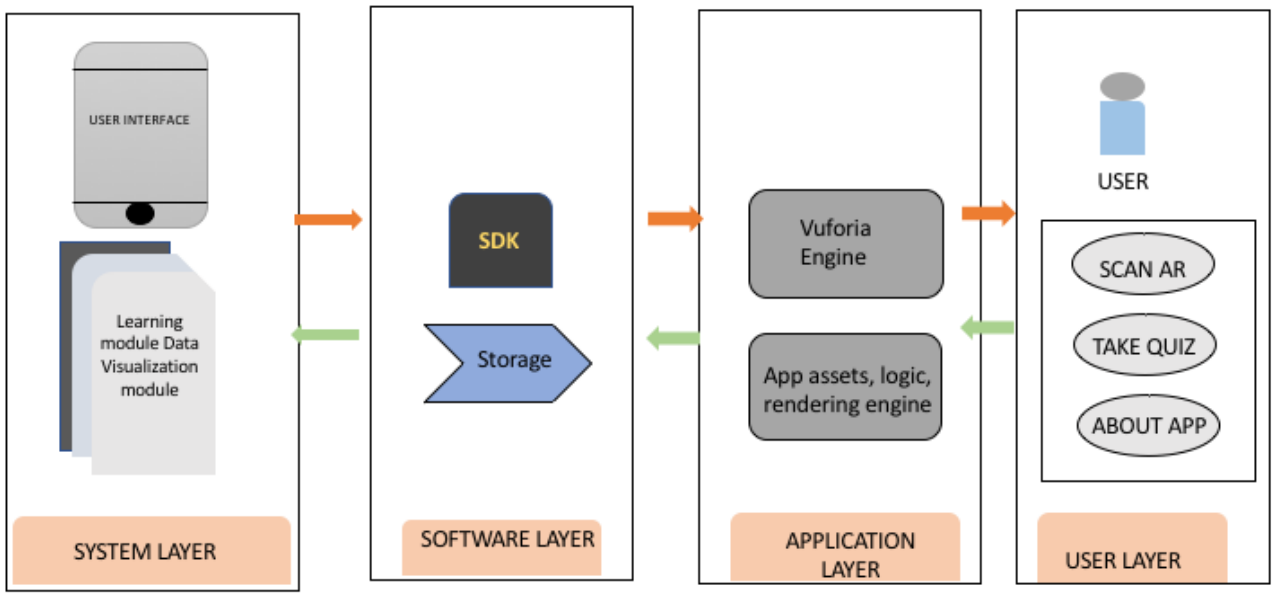


Figure 3.2 Architecture Design

Figure 3.2 shows the architecture design of the prototype. It consisted of 3 layers which are features of the system application which are Scan AR, Take Quiz, and About app. Scan marker is where the user will capture the text-based content from the learning module and proceed to the detection of marker. Other than that, about features are where users can get details information about data visualization subject content.

Next, this system application is also connected to software development kits (AR SDK) which provide more digital content and it is also connected between the system and mobile interface. There is also a Vuforia engine that works for augmented reality cameras and databases.

The shape of the rectangular is the mobile interface of this design. These elements are user, input-output devices, user interface, and process. The user(s) of the prototype is an analyst and he/she uses input-output devices. The learning module is where the learning text-based for a user to do the scanning.

3.3.2 Use Case Diagrams

In the prototyping phase, the design overall project will be developed based on the overall use case diagram shown in **Figure 3.3**

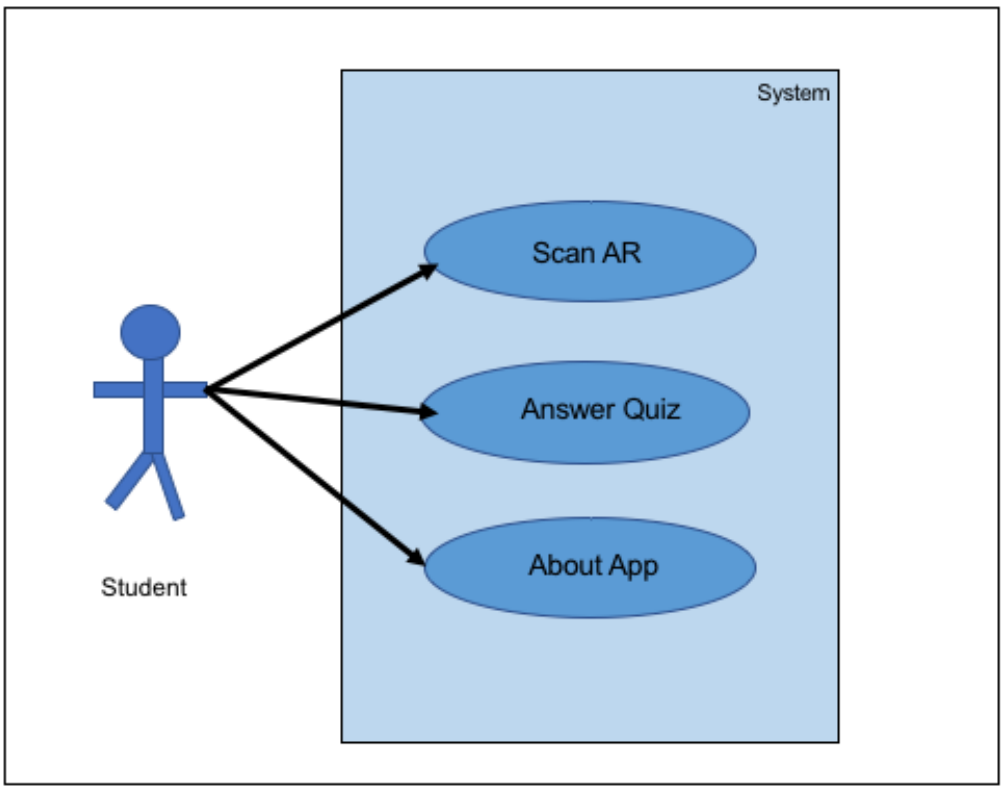


Figure 3.3 Overall use case diagrams

Actors: Students that take DSV650 subject

The actor in this system is a user where the user interacts with the function of the system. There are two use cases in the diagram that include the functionality of the system which are Scan AR, Answer Quiz and About App. When users running and open the system on the mobile application, the system will display the main page with these buttons. Each button will navigate the user to another page of the system and this is where the user can know the function of the system, on what the system does.

3.33 Description Use case 1: Scan AR

Table 3.2 Description of Scan AR

1a. Id No.	UD1	**1b. Title**	UC1- Scan AR
2. Actor	1.Student 2.User		
3. Brief Description	1. This use case is a feature for the user to scan the content-base from data visualization textbook before viewing the 3D modeling		
4. Precondition	User must be students that take data visualization subject		
5. Normal Flow	1. This use case starts when the system display camera for the user to do the scanning. 2. Users choose any image in the data visualization textbook for 3d modeling objects to display. 3. The user pointing the camera right at the particular image in the learning module for scanning. The device must first recognize which image do user point to. 1.1 If the image that the user point is not detected by the camera, please refer to Exception Flow 1. 4. The picture will be recognized and 3D modeling starts to appear modeling and user able to interact with the model by viewing information. 5. The use case ends.		
6. Extension – Alternate Flow			
7. Extension – Exception Flow	1. Error Message 1 1.1 The system will display the error message "Undetected target image, please point it closer". 1.2 This use case continues at Normal Flow 1.		

3.3.4 Activity Diagram for Scan AR

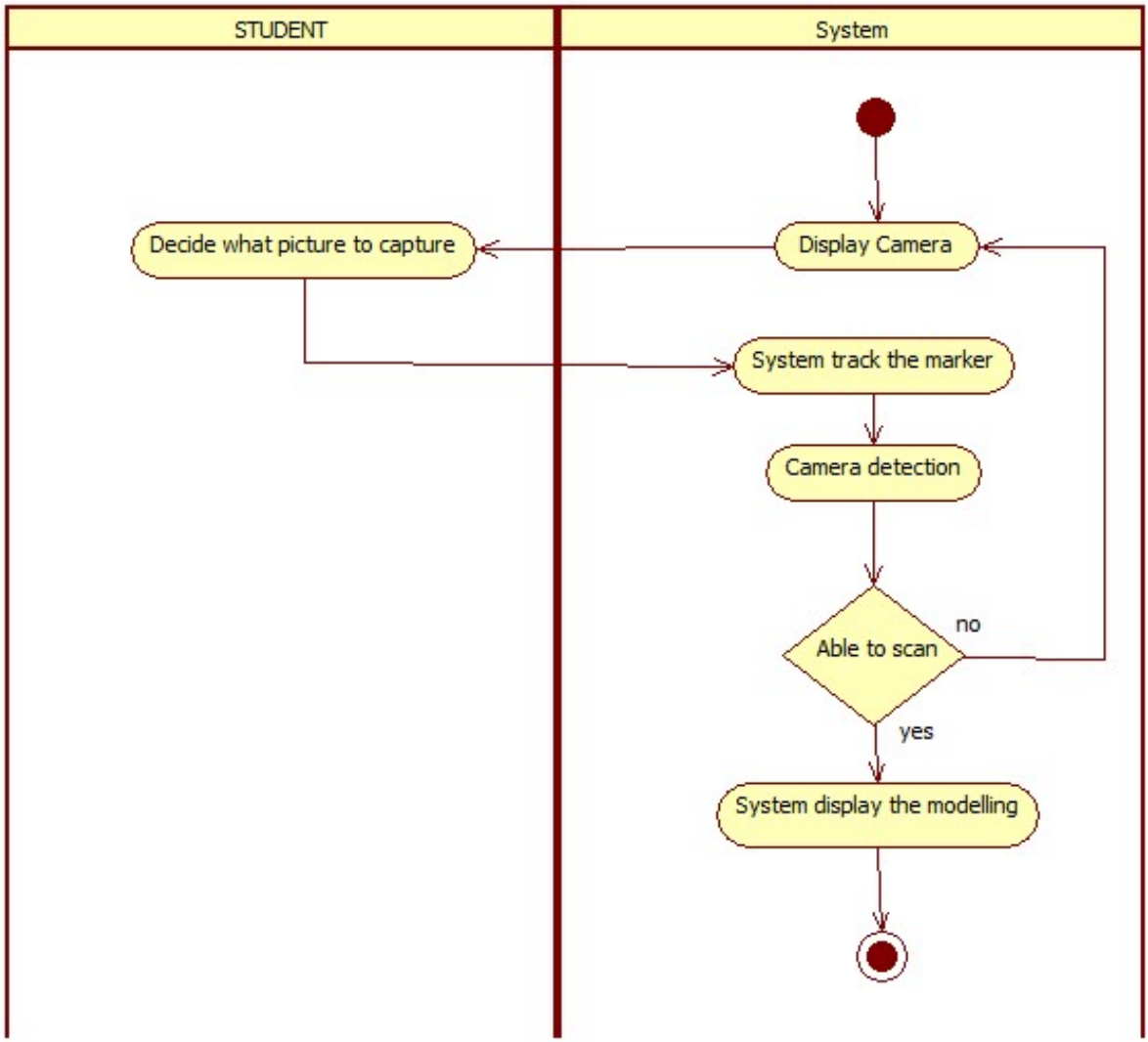


Figure 3.4 Activity diagram Scan AR

3.3.5 Description Use case 2: Answer Quiz

3.2 Description of taking Quiz

1a. Id No.	UD2	**1b. Title**	UC1-Take Quiz
2. Actor	1.Student 2.User		
3. Brief Description	1. This use-case will measure a user's understanding by answering a few questions after they had viewing 3D-modelling object.		
4. Precondition	The user must answer all the question and marks will be given		
5. Normal Flow	1. This use-case begins with objective questions and the user must choose only one correct answer. 1.1 If the user did not finish answering all the questions, please refer to Exception Flow 1. 2. Users submit the answer and the system will display the marks for the correct answer that the user submitted. 3. The use-case ends		
6. Extension – Alternate Flow	1. Answering Quizzes question 1.1 Users can skip/hold to any question that they want to answer first or question that they don't understand.		
7. Extension – Exception Flow	2. Error message 2.1 The system will display an error message "Oops! Looks like there is a question you left behind". 2.2 This use case continues at Normal Flow 1.		

3.3.6 Activity Diagram for Take Quiz

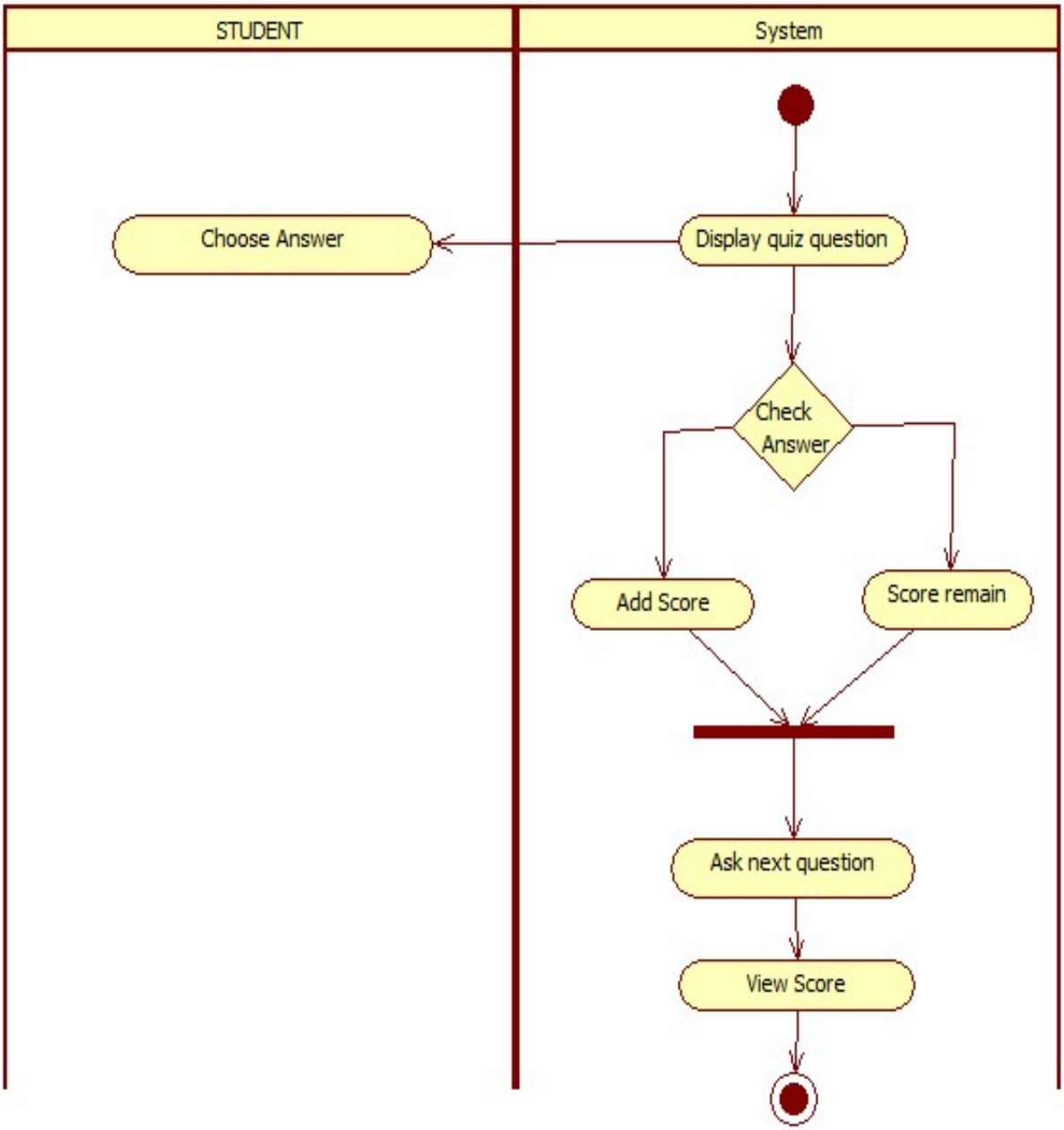


Figure 3.5 Activity diagram Answer Quiz

3.3.7 Sequence Diagram for Scan AR

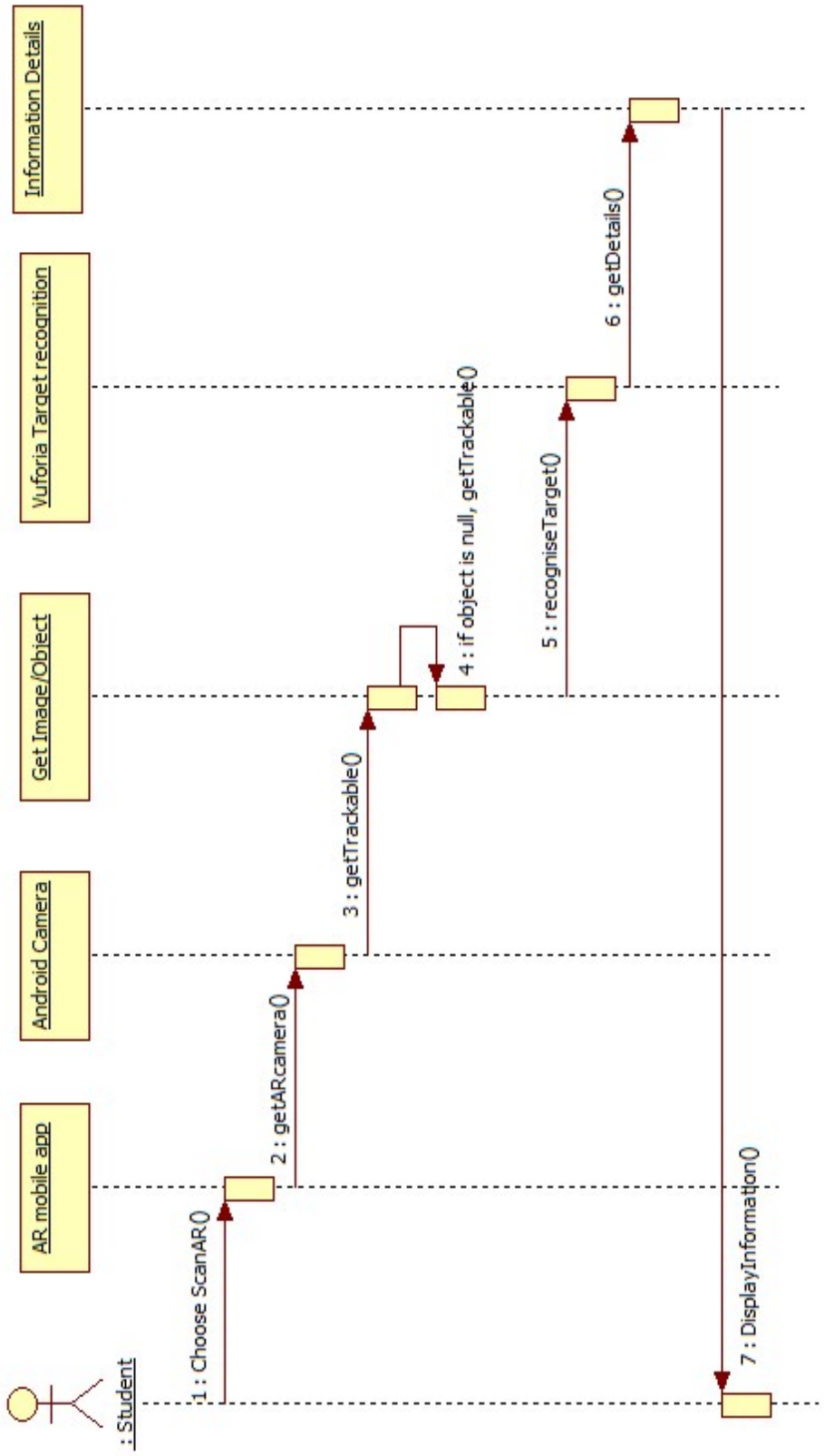
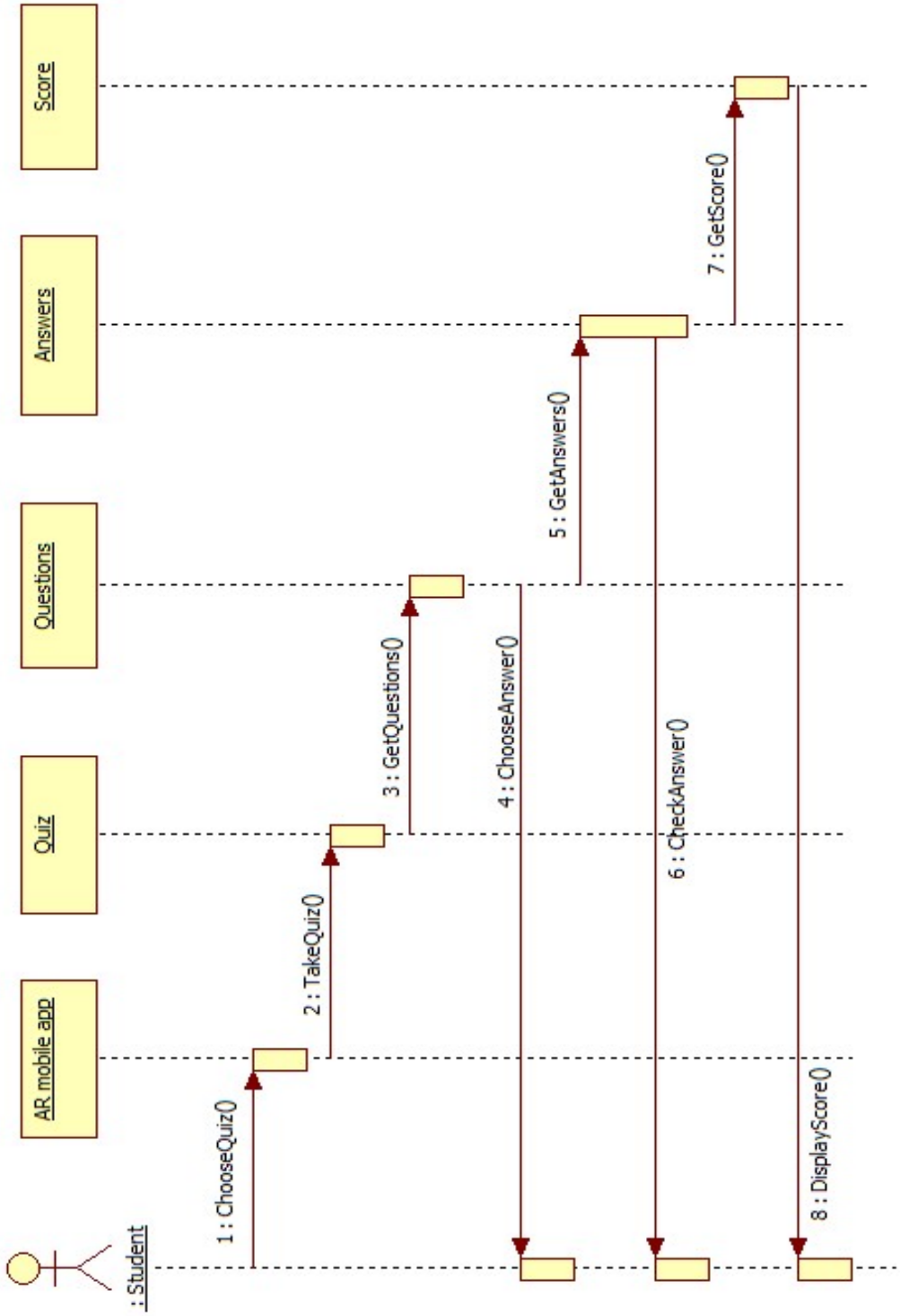


Figure 3.6 Sequence diagram for Scan AR

3.3.8 Sequence Diagram for Take Quiz



3.7 Sequence Diagram for Take Quiz

3.4 Testing

After the development phase, the system needs to be tested to verify the system's functionalities and ensure it performs as desired. There are two main parts of testing which are functionality testing and user testing. Functionality testing is conducted by testing each function independently and the integration between these functions. In user testing, a suitable person is chosen and the prototype is delivered. The user uses his/her data and carries out interactions. Then the feedback is sent to the developer.

3.5 Software and Hardware Requirement

Software and hardware requirement is the most important things needed to develop the system. The sophistication of the software and hardware allows the developer to develop the system effectively and systematically. Table 3.14 and 3.15 shows the list of software and hardware to develop this system.

Table 3.16 List of software

No	Software	Specification
1	Adobe Illustrator	Version 2020
2	Unity	Version 2018
3	Vuforia	Version 8.9
	Microsoft office	Version 2016
4	Mendeley	Version 1.6.1

Table 3.17 List of hardware

No	Hardware	Specification
1	System Unit	Intel Core i5 processor RAM 4GB HDD 500GB GEFORCE GTX 1050
	Smartphone	Android

CHAPTER 4

DESIGN AND DEVELOPMENT

This chapter is about the overall design of the prototype and development phase of Learning Data Visualization Subject using an Augmented Reality based system that has been discussed in the previous chapter of this report.

4.1 Development Overview

In this phase, the development of the mobile application system will be developed and the function of the system must be well defined. Table 4.1 shows the overall main function of this mobile application system.

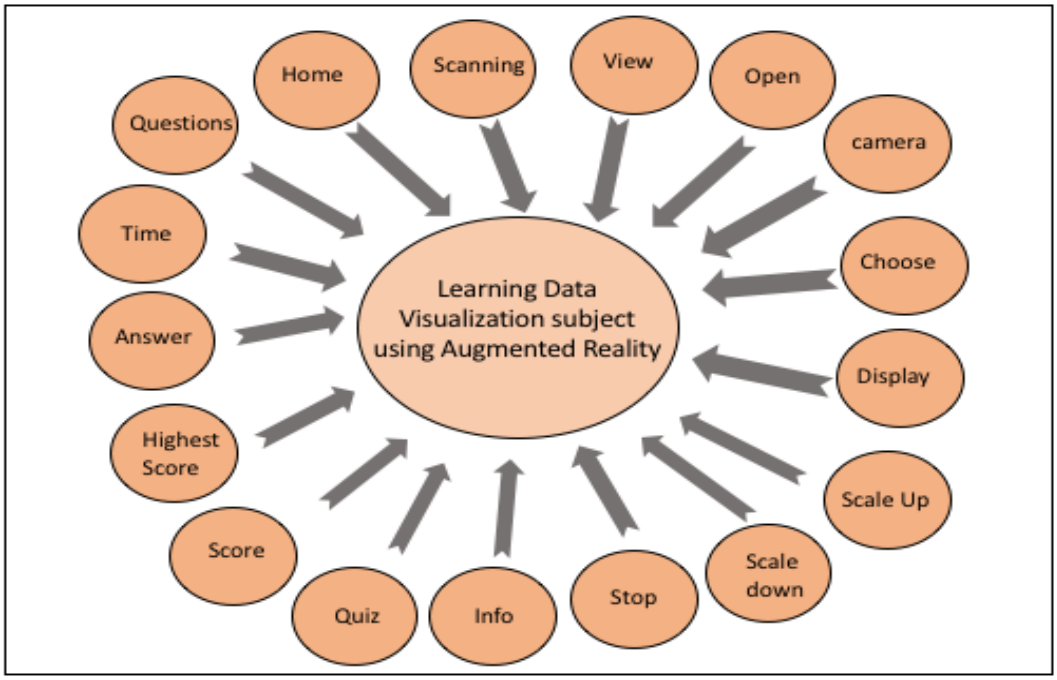


Figure 4.1 Overall main function

Table 4.1 Description of the overall main function

Function	**Description**
Open	To use this system, a user must click the toggle to open the main menu
Camera	To view the modeling object, a user must capture the object through the camera function
Choose marker	Allow user to choose which marker they want to scan
Scanning marker	To display the object modeling, user need to scan marker for detection
Display object modeling	User will able to see object modeling display in the screen
Display video	User will be able to see object played in the video
Display animation	User will be able to see object played in the animation
View information	Allow users to view information about the modeling in detail.
Scale Up object	Allow user to scale up the object modeling
Scale Down object	Allow user to scale down the object modeling
Stop Scaling	Allow users to stop scaling object modeling.
Information about the application	This function is for the first time user to know how to use the system and how the system works.

Quiz	Allow students to enhance their understanding by taking the quiz
Score	This function is to display the score that the user gets after answering quiz
Highest Score	This function is to display the highest score get from the previous user
Answer	User need to choose one correct answer only
Time	This function display time for the user to answer all the questions
Quiz Questions	This function display questions for students to answer related to what they learned
Home	This function is to navigate the user to the main menu

4.2 Class Diagram

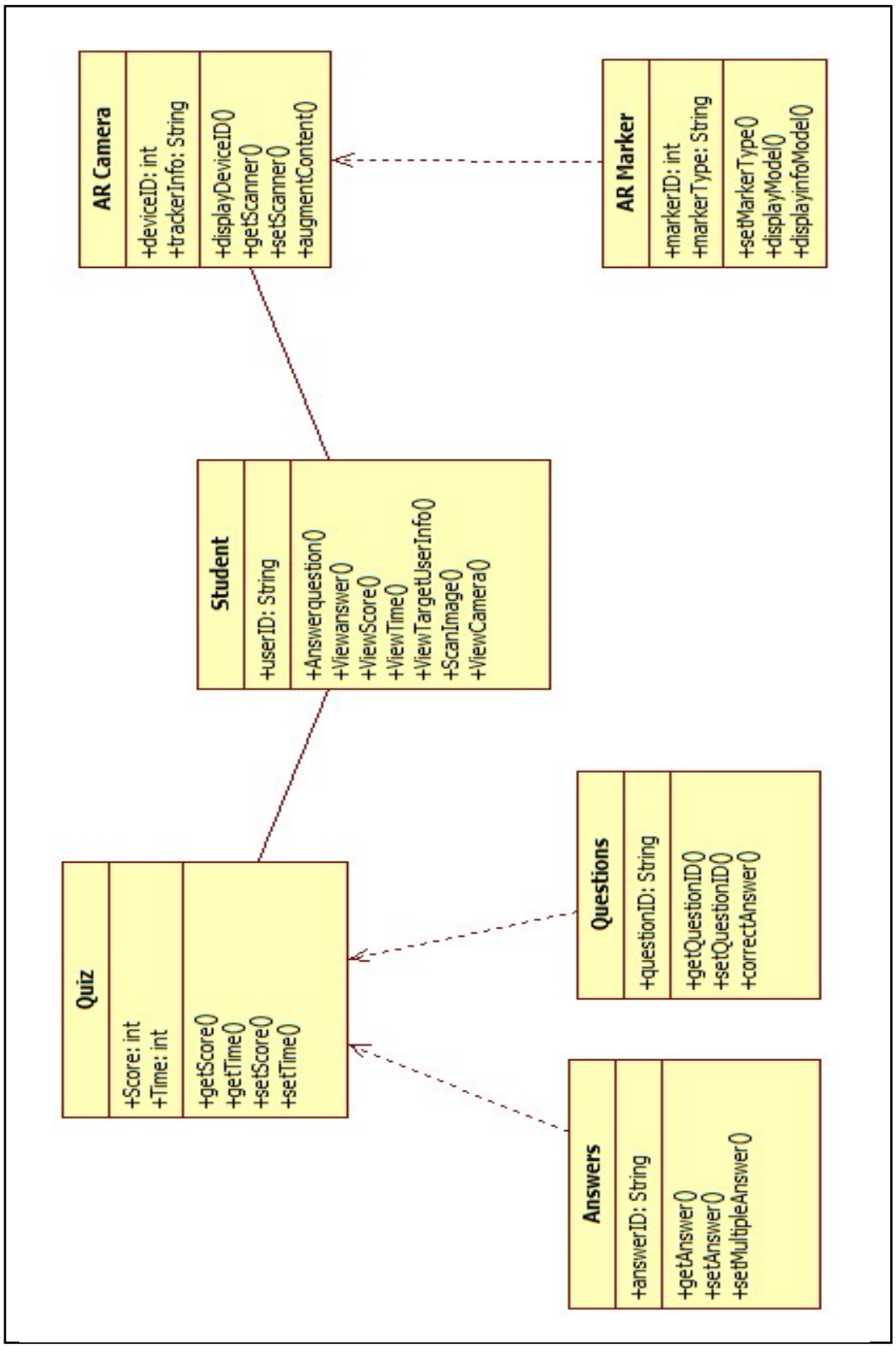


Figure 4.12 Class Diagram

4.3 Overall System Flow

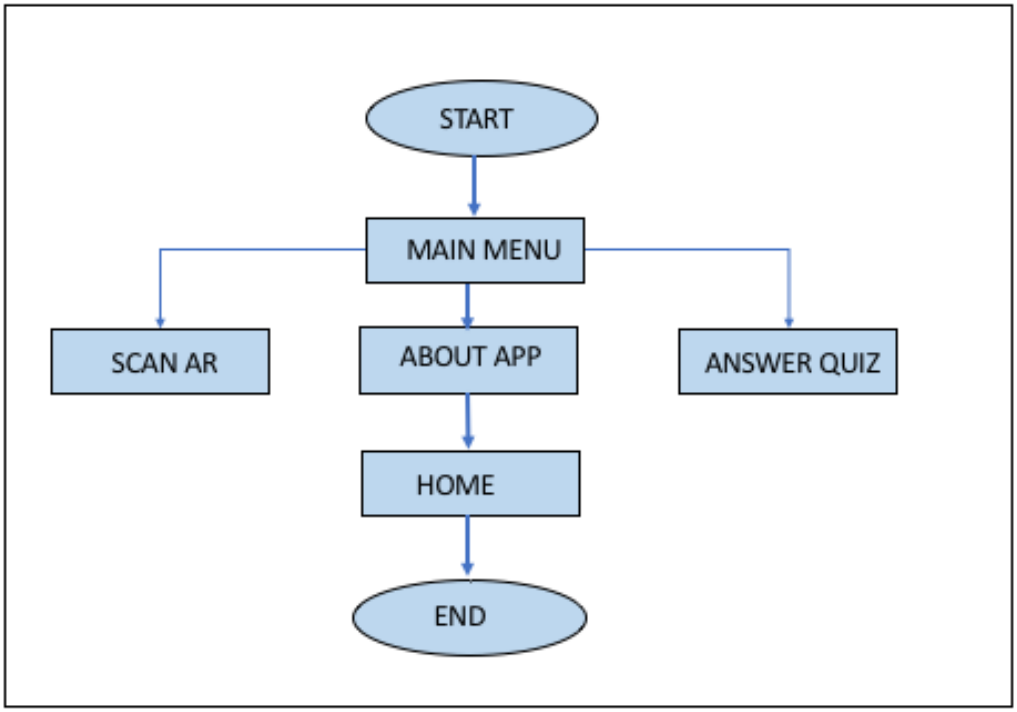


Figure 4.13 Overall system flow

Table 4.2 Overall system flow description

Process	**Description**
Main Menu	The user will be automatically directed to the main menu. In the main menu, it will show buttons for the user to navigate.
Scan AR	This page will enable the user to scan the marker from the learning module and able to view object modeling
Answer Quiz	This page is for the user to gain their understanding of the subject by answering quiz
About App	This Page enables a user to view and know details about the system
Home	This feature is to navigate the user back to the main menu

4.3.1 Student System Flow

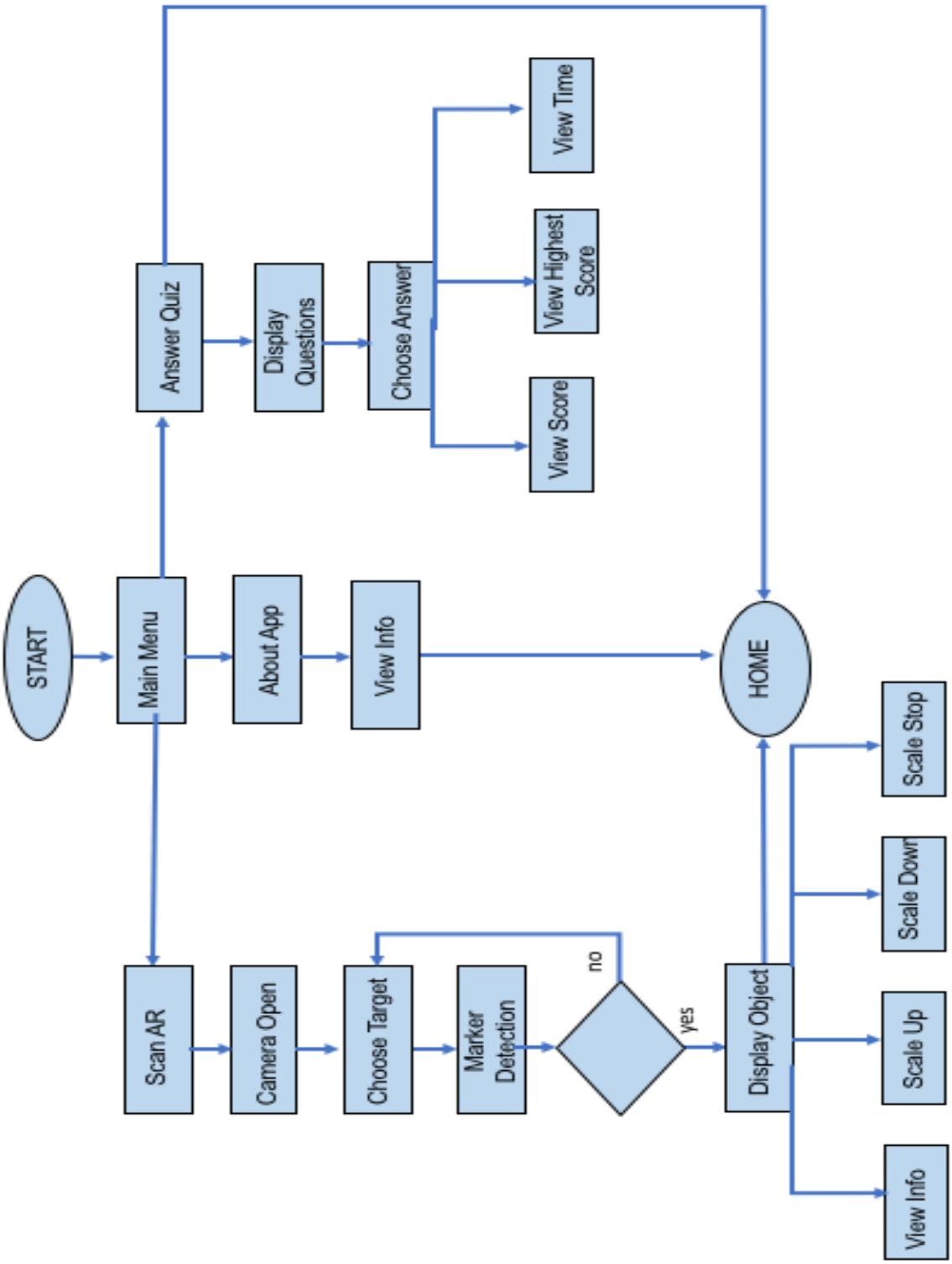


Figure 4.14 Student System Flow

Table 4.3 Student System Flow Description

Process	**Description**
Main Menu	The user will be automatically directed to the main menu. In the main menu, it will show buttons for the user to navigate.
Scan AR	This page will enable the user to scan the marker from the learning module and able to view object modeling
Answer Quiz	This page is for a user to gain their understanding of the subject by answering quiz
About App	This page enables the user to view and know details about the system
Home	This feature is to navigate the user back to the main menu

4.4 Process

The process is an action of a program that runs on a system. Each function in the system shows the flow of the system works and it is important to make a process clear.

4.4.1 Scan AR process

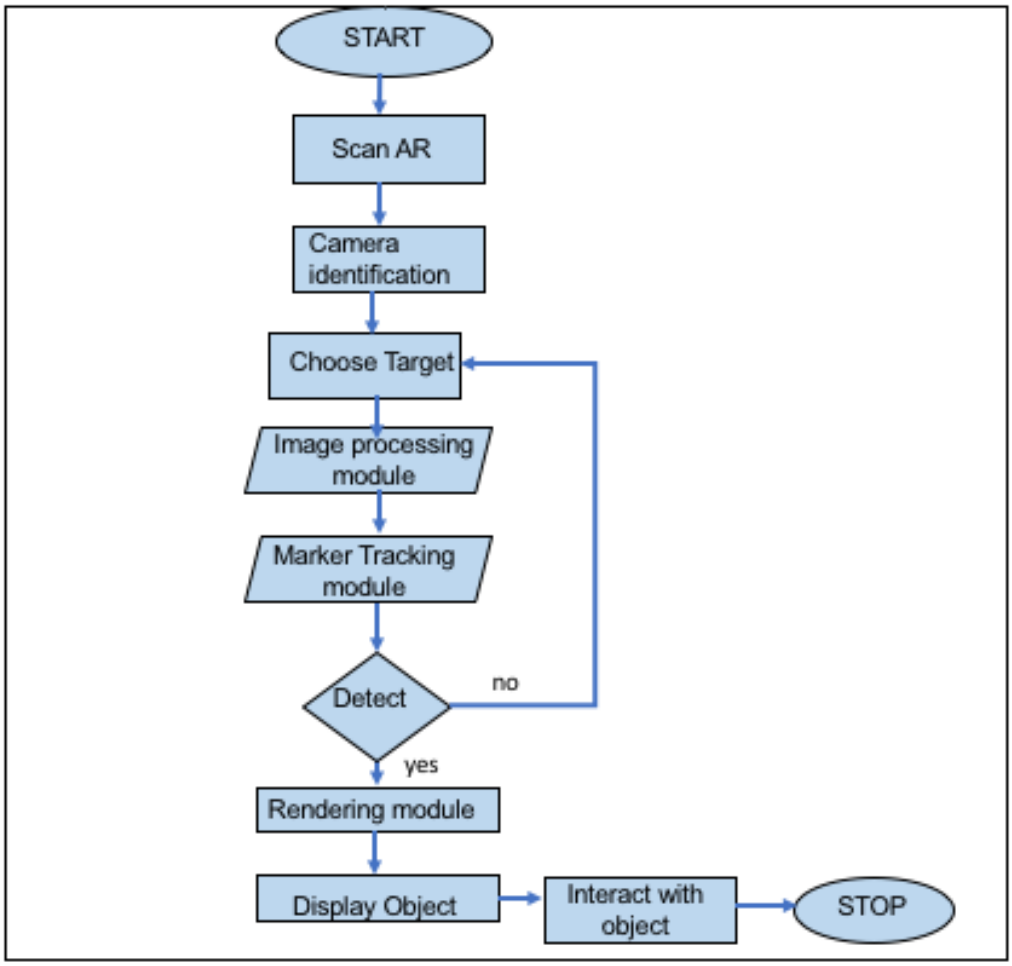


Figure 4.15 Scan AR system flowchart

4.4.2 Image processing module in Scan AR

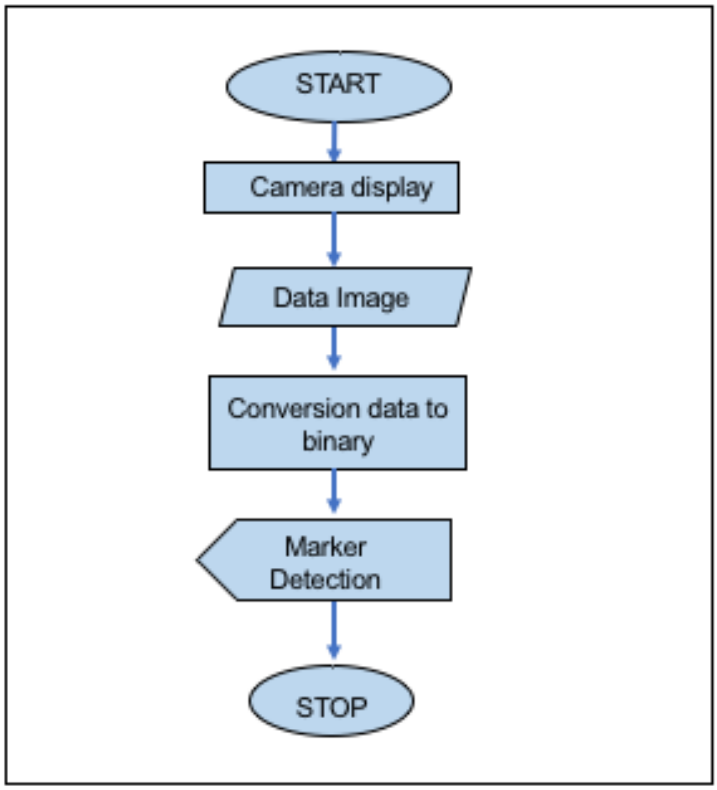


Figure 4.16 Image processing module in Scan AR system flow

4.4.3 Marker Tracking module in Scan AR

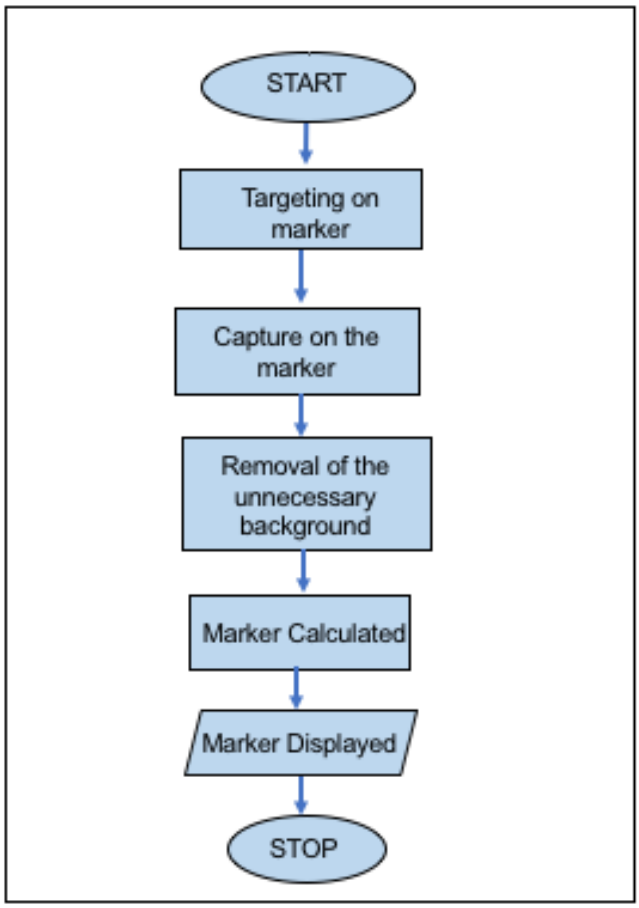


Figure 4.17 Marker Tracking module in Scan AR system flow

4.4.4 Take Quiz process

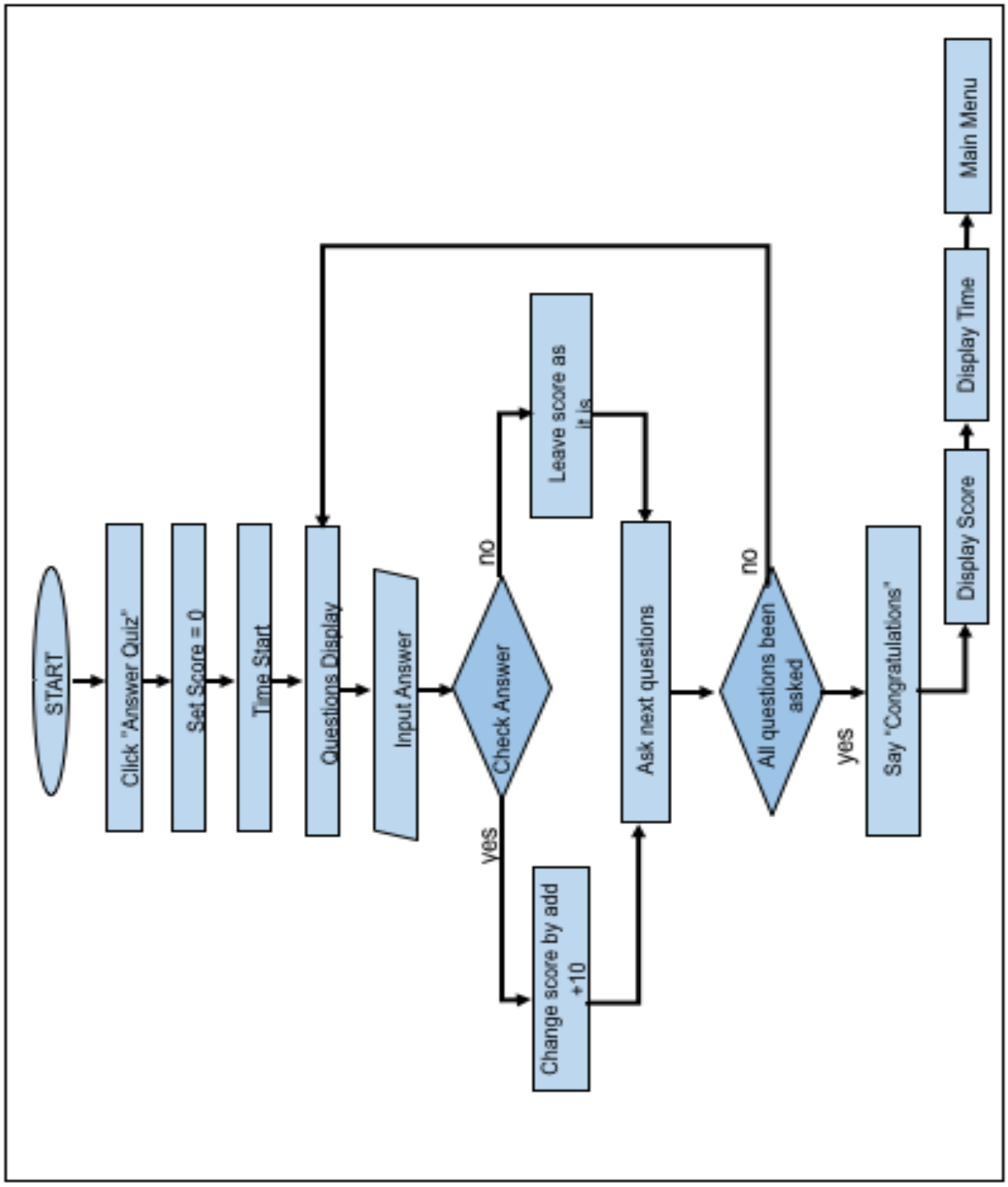


Figure 4.18 Take Quiz system flow

4.4.5 View Info process

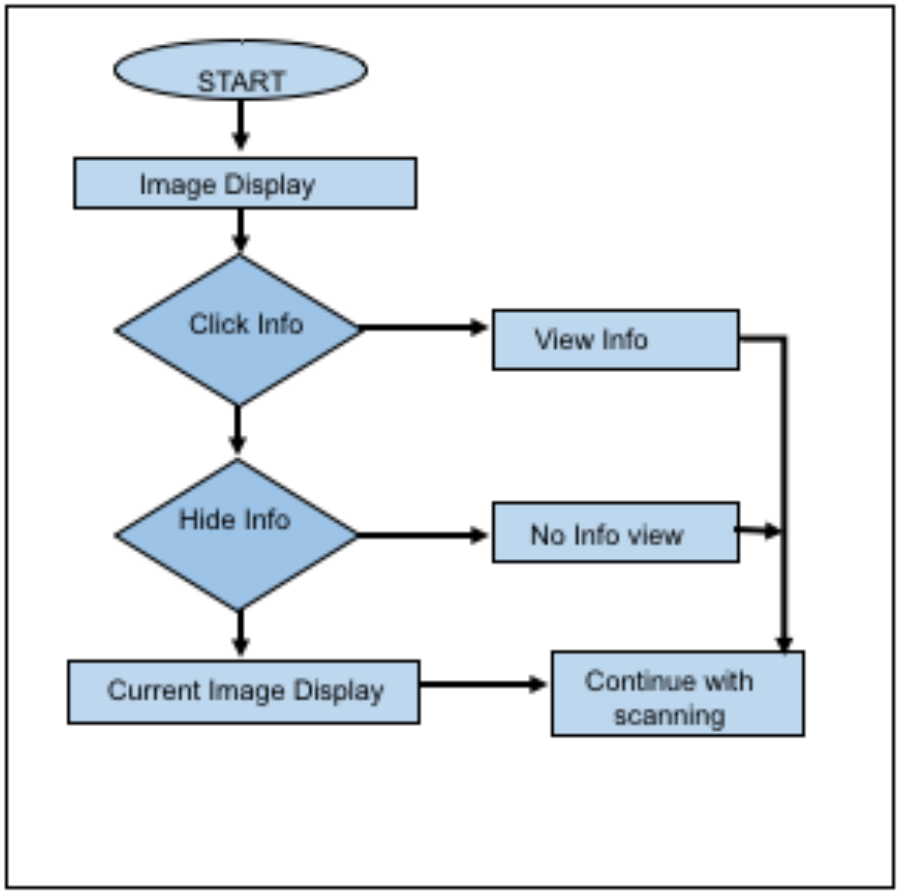


Figure 4.19 View Info system flow

4.4.6 Image Display Process

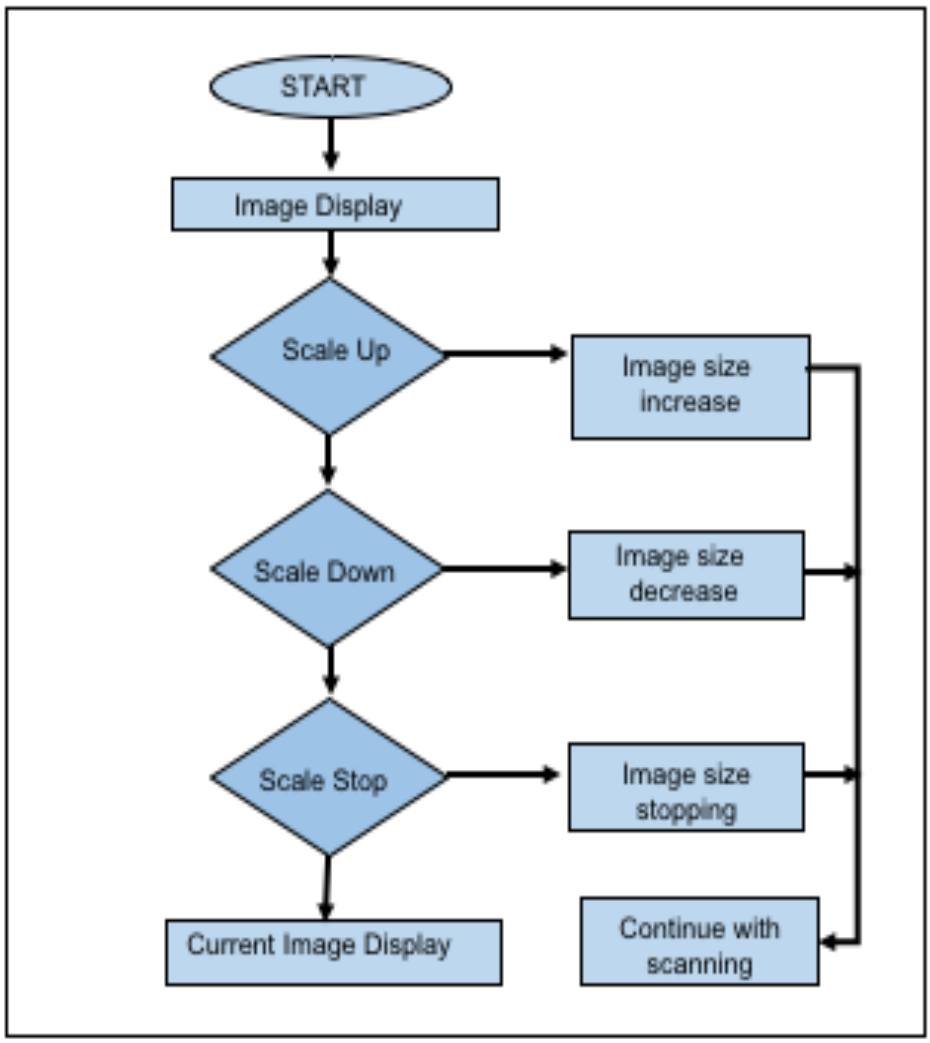


Figure 4.20 Output for Image Display system flow

4.7 System Procedure

System procedure is a fixed-activities that need to be performed step by step. To perform a correct task, the procedure is important to ensure that all the standard step that has been determining are well performed

4.7.1 Image Processing Module procedure

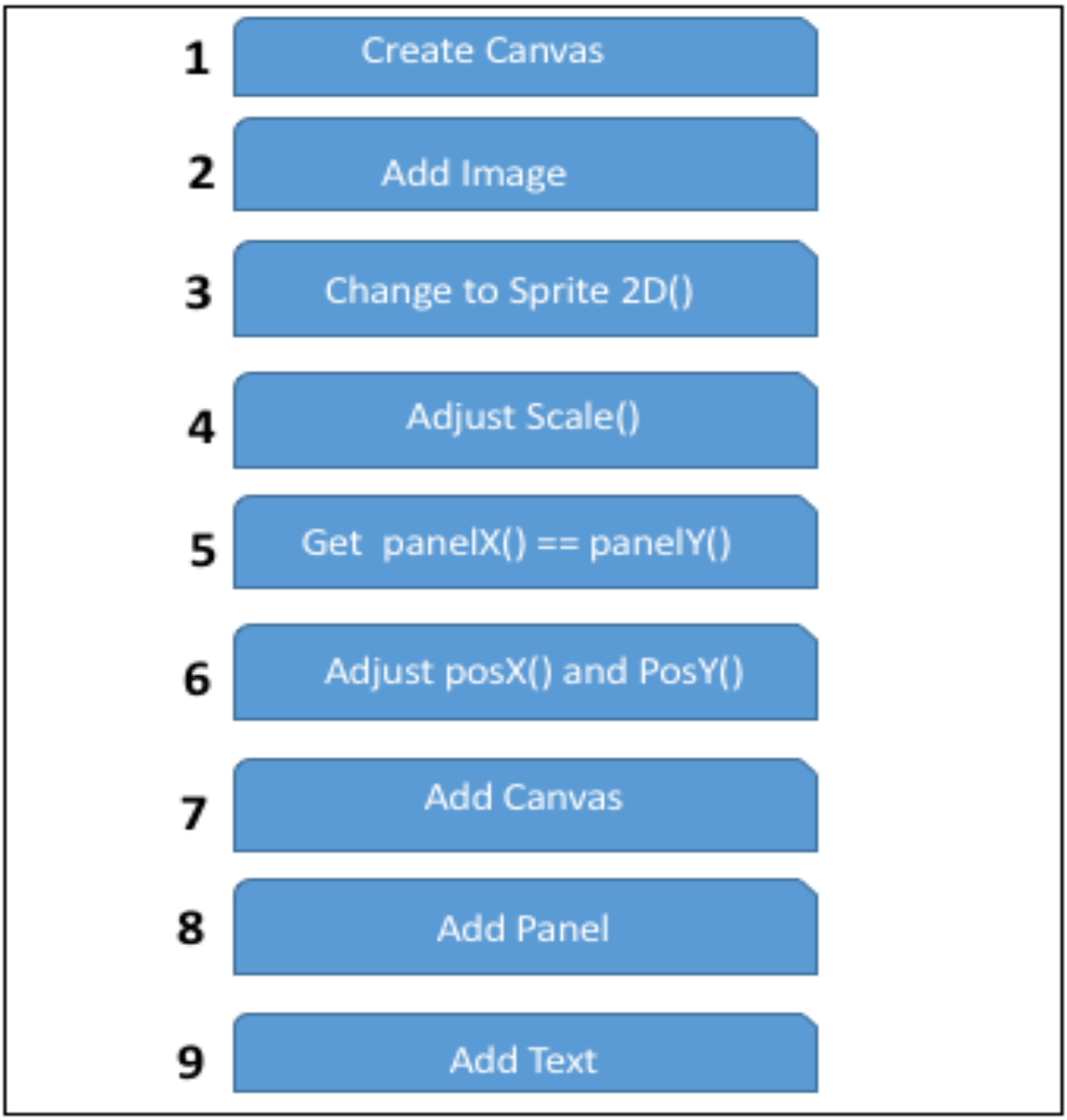


Figure 4.21 Image Processing module

There are nine steps in developing the Image module as shown in Figure above. The first step is to create an empty canvas. The canvas works like a frame for the image to detect and display. In the second step. As the original image that has been added is a set to default image, this image needs to be converted to sprite 2D to make the image detection more accurate which happens in the third step. In the fourth step, the image is the scale and adjust accordingly to the size of the canvas. In the fifth step, the value of panel X and panel Y between the canvas and image must be in the same width and height. In the sixth step, pos X and pos Y that represent position must be in the same position between the image and the AR camera. In the seventh step, another canvas is added to work as a frame used to view information. In the eighth step, the panel is added to display text. The last step is to add text for the information display.

4.7.2 Marker-Tracker procedure

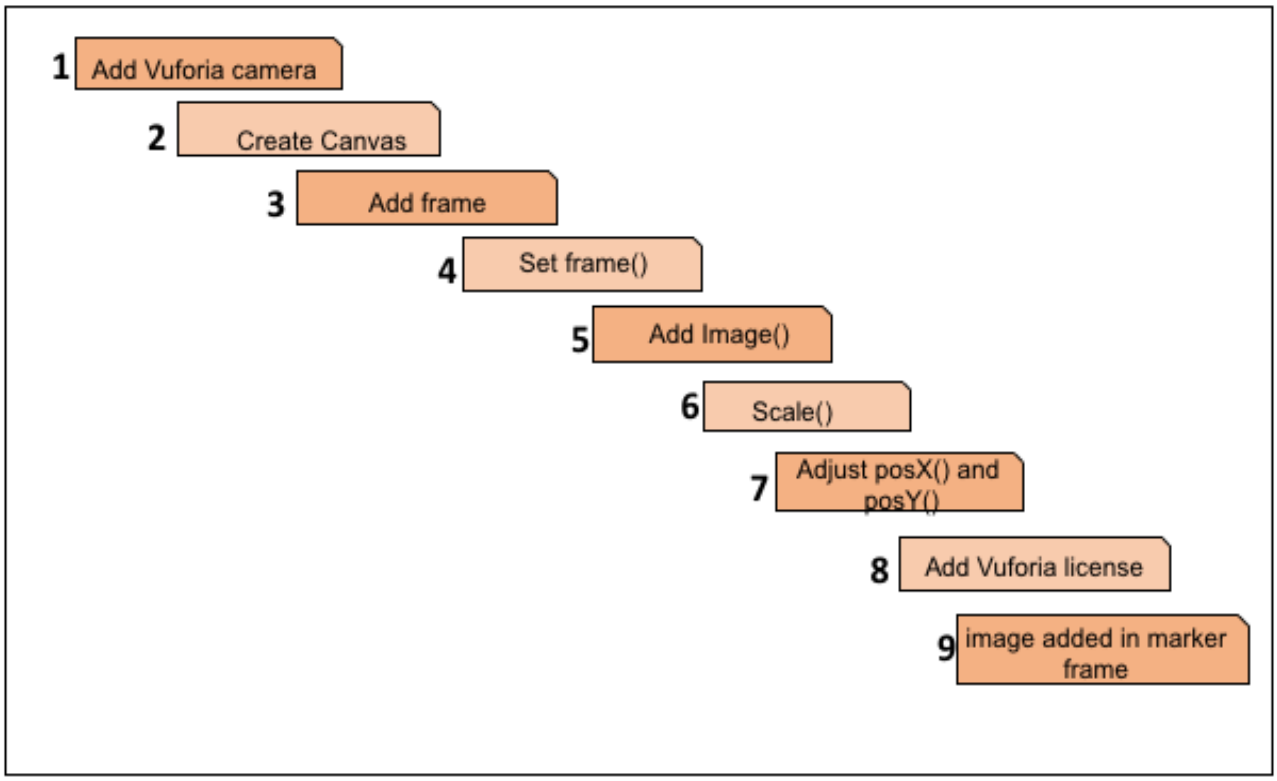


Figure 4.22 Marker-Tracker

There are nine steps in developing Marker Tracker as shown in Figure above. The first step is to import and add a Vuforia camera. Vuforia camera functions as a camera tracker to capture the marker. The second step is to create a canvas that works like a frame for the image to detect and display. In the third and fourth steps, the frame is added and set accordingly to the camera target. The fifth step is to add an image as a marker. In sixth and seventh steps is to scale the image and adjust for X and Y position to fit in the camera frame. Eighth is to add the Vuforia license key to able the Vuforia camera function. In the last step, an image is added to the marker frame, and images are set as a marker. Figure 4.20 and Figure 4.21 shows the segment of code of marker tracker.

```
18      public class DefaultTrackableEventHandler : MonoBehaviour, ITrackableEventHandler
19      {
20
21
22
23
24          #region PROTECTED_MEMBER_VARIABLES
25
26          protected TrackableBehaviour mTrackableBehaviour;
27          protected TrackableBehaviour.Status m_PreviousStatus;
28          protected TrackableBehaviour.Status m_NewStatus;
29
30          #endregion // PROTECTED_MEMBER_VARIABLES
31
32          #region UNITY_MONOBEHAVIOUR_METHODS
33
34          protected virtual void Start()
35          {
36              mTrackableBehaviour = GetComponent<TrackableBehaviour>();
37              if (mTrackableBehaviour)
38                  mTrackableBehaviour.RegisterTrackableEventHandler(this);
39          }
40
41          protected virtual void OnDestroy()
42          {
43              if (mTrackableBehaviour)
44                  mTrackableBehaviour.UnregisterTrackableEventHandler(this);
45          }
46
47          #endregion // UNITY_MONOBEHAVIOUR_METHODS
48
49          #region PUBLIC_METHODS
50
51          /// <summary>
52          ///     Implementation of the ITrackableEventHandler function called when the
53          ///     tracking state changes.
54          /// </summary>
55          public void OnTrackableStateChanged(
56              TrackableBehaviour.Status previousStatus,
57              TrackableBehaviour.Status newStatus)
58          {
59              m_PreviousStatus = previousStatus;
60              m_NewStatus = newStatus;
61
62              if (newStatus == TrackableBehaviour.Status.DETECTED ||
63                  newStatus == TrackableBehaviour.Status.TRACKED ||
64                  newStatus == TrackableBehaviour.Status.EXTENDED_TRACKED)
65              {
66                  Debug.Log("Trackable " + mTrackableBehaviour.TrackableName + " found");
67                  OnTrackingFound();
68              }
69              else if (previousStatus == TrackableBehaviour.Status.TRACKED &&
70                       newStatus == TrackableBehaviour.Status.NO_POSE)
71              {
72                  Debug.Log("Trackable " + mTrackableBehaviour.TrackableName + " lost");
73                  OnTrackingLost();
74              }
75              else
76              {
77                  // For combo of previousStatus=UNKNOWN + newStatus=UNKNOWN|NOT_FOUND
78                  // Vuforia is starting, but tracking has not been lost or found yet
79                  // Call OnTrackingLost() to hide the augmentations
80                  OnTrackingLost();
81              }
82          }
83
84          #endregion // PUBLIC_METHODS
85
86          #region PROTECTED_METHODS
87
88          protected virtual void OnTrackingFound()
89          {
90
91              var rendererComponents = GetComponentsInChildren<Renderer>(true);
92              var colliderComponents = GetComponentsInChildren<Collider>(true);
93              var canvasComponents = GetComponentsInChildren<Canvas>(true);
94
95
96              // Enable rendering:
97              foreach (var component in rendererComponents)
98                  component.enabled = true;
99
100             // Enable colliders:
101             foreach (var component in colliderComponents)
102                 component.enabled = true;
```

Figure 4.23 Segments code of marker- tracker

4.7.3 Creating Interface for toggle button

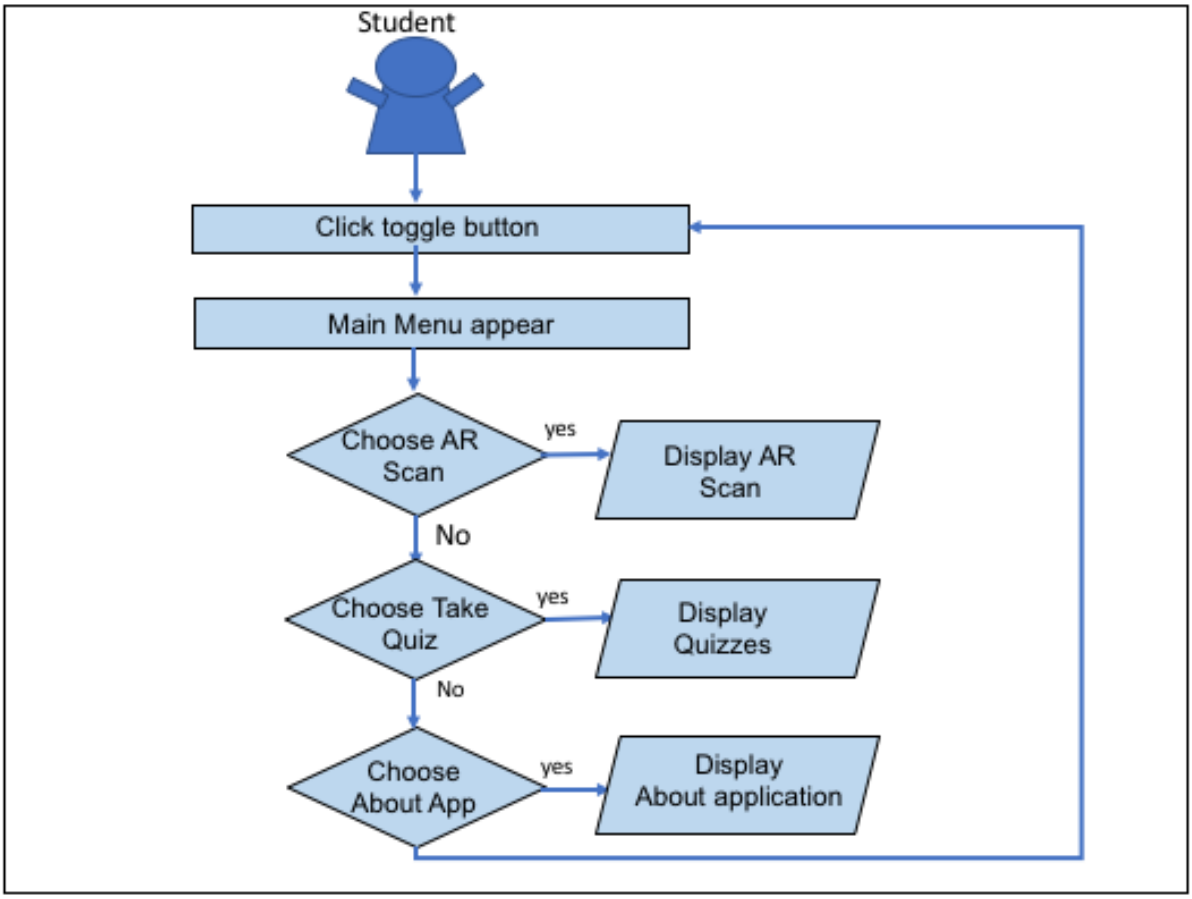


Figure 4.24 Open Interface using toggle button

Users will click the toggle button to open the main menu. In the main menu, there will be three choices of buttons that the user can choose to view. Figure 4.26 and 4.28 shows the segment code and interface of the main menu with the toggle button.

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class SliderMenu : MonoBehaviour
{
    public GameObject PanelMenu;

    public void ShowHideMenu()
    {
        if (PanelMenu != null)
        {
            Animator animator = PanelMenu.GetComponent<Animator>();
            if (animator != null)
            {
                bool isOpen = animator.GetBool("show");
                animator.SetBool("show", !isOpen);
            }

        }
    }
}
```

Figure 4.25 Segments of code for the toggle button

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.UI;

public class AdjustImage : MonoBehaviour
{
    public float rotationMin = 0.0f;
    public float rotationMax = 45.0f;
    public Slider RotateSlider;
    public Slider ScaleSlider;

    [SerializeField] float currentRotation = 0f;
    [SerializeField] float currentScale = 0.1f;

    void Start()
    {
        RotateSlider = GameObject.Find("RotateSlider").GetComponent<Slider>();
        ScaleSlider = GameObject.Find("ScalesSlider").GetComponent<Slider>();
    }

    void Update()
    {
        transform.localEulerAngles = new Vector3(0.0f, RotateSlider.value,0.0f);
        transform.localScale = new Vector3(ScaleSlider.value, ScaleSlider.value,ScaleSlider.value);
    }
```

Figure 4.26 Segments code to animate toggle button

```
}

private void OnGUI()
{
    currentRotation = GUI.HorizontalSlider(new Rect(-280f,165.0f,228.0f,57.0f),currentRotation,0.0f,45.0f);
    transform.localEulerAngles = new Vector3(0.0f, currentRotation, 0.0f);
    currentRotation = GUI.HorizontalSlider(new Rect(-280f, 120.0f, 228.0f,57.0f), currentScale, 0.0f, 2f);
    transform.localScale = new Vector3(currentScale,currentScale,currentScale);

}

public void AdjustAngle(float newAngle)
{

}
public void AdjustScale(float newScale)
{

}
}
```

Figure 4.27 Segments code for User Interface

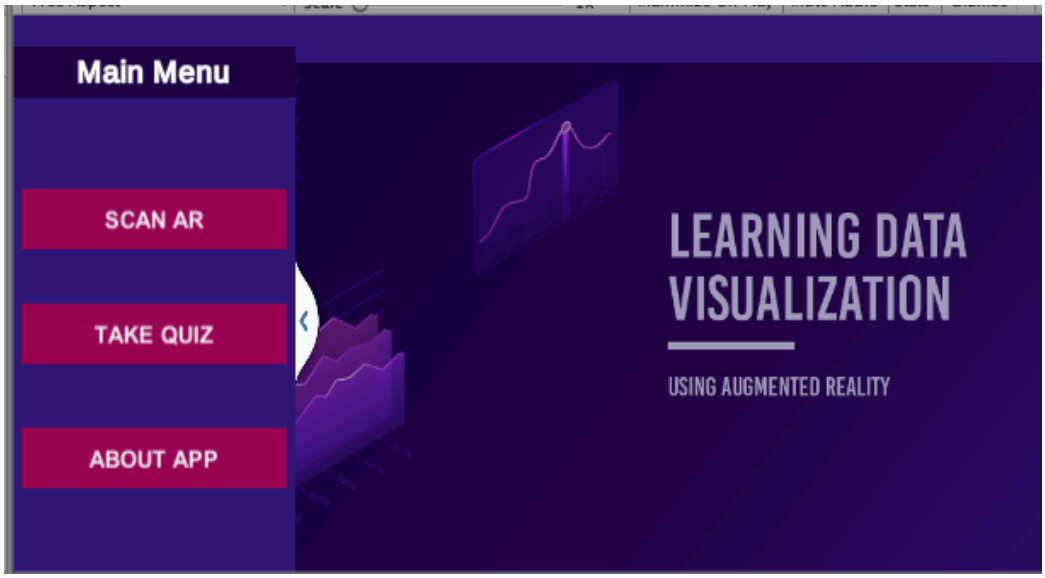


Figure 4.28 User interface of the system

4.7.3 Creating Info view in object modeling Scan AR

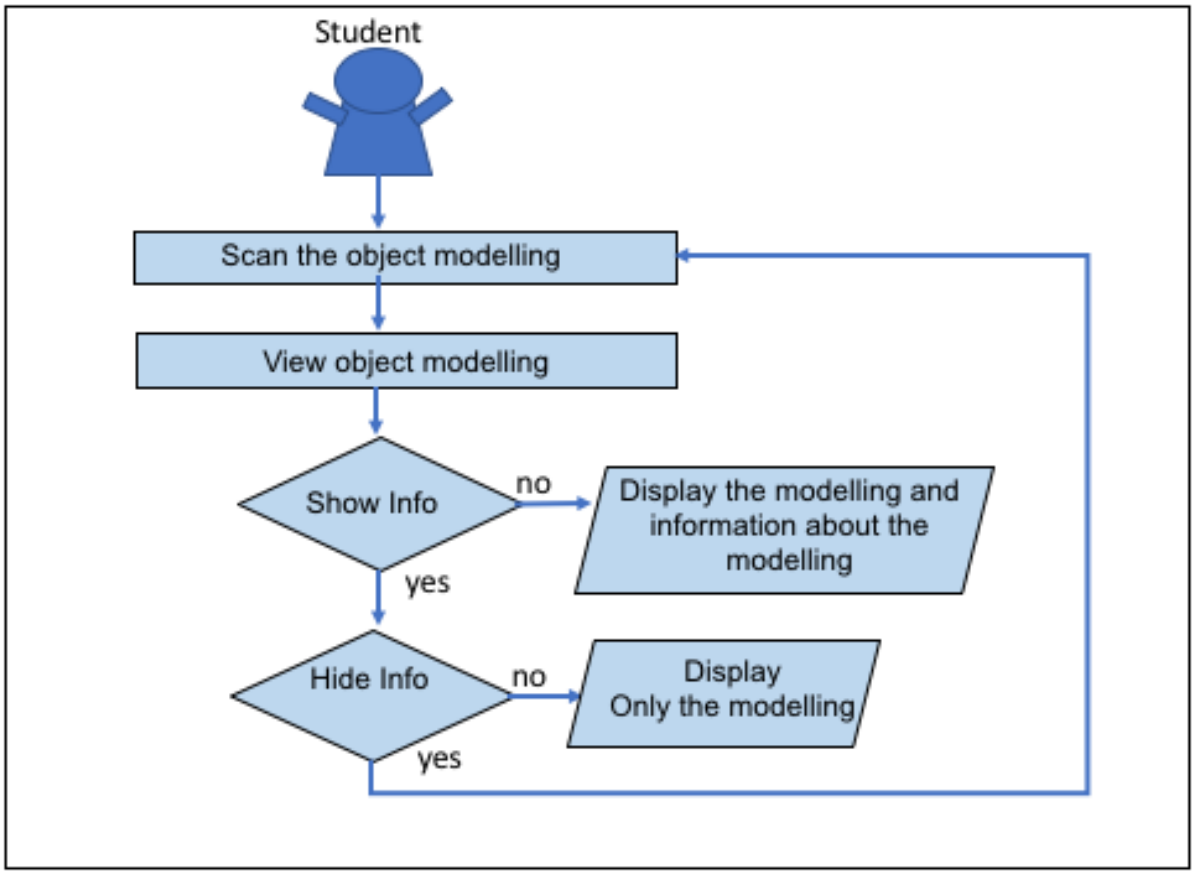


Figure 4.29 View information from object modeling

When a user is viewing the object modeling, the user can click "read more" to view information from the object modeling display. Hide info is when the user only wants to view the object modeling. **Figure 4.30** shows the segment code for the information display button.

No selection

```
1   using System.Collections;
2   using System.Collections.Generic;
3   using UnityEngine;
4
5   public class Controller : MonoBehaviour
6   {
7       public GameObject PanelInfo;
8
9       private bool show = false;
10
11      public void ShowHideInfo()
12      {
13          if (!show)
14          {
15              PanelInfo.SetActive(true);
16              show = true;
17          }
18          else
19
20          {
21              PanelInfo.SetActive(false);
22              show = false;
23
24          }
25      }
26  }
```

Figure 4.30 shows the segment of code for show and hide info

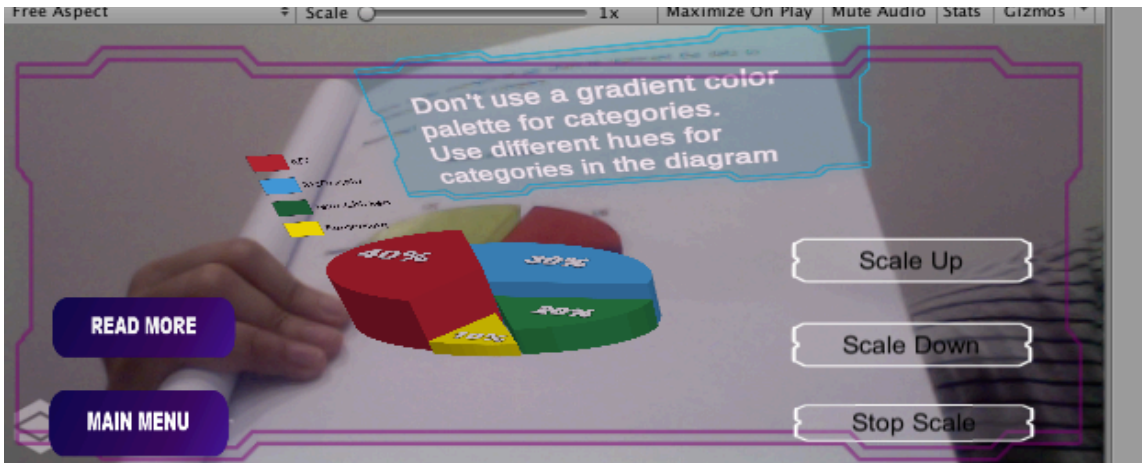
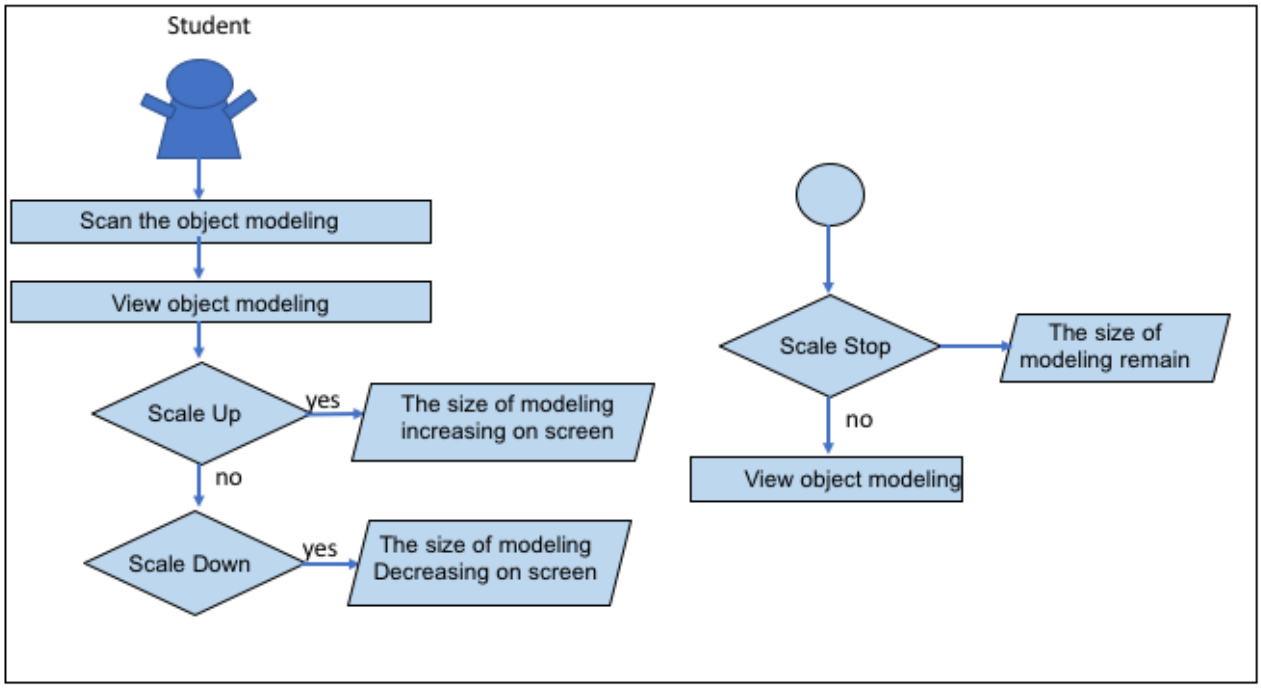


Figure 4.31 Example for output Info

4.7.4 Creating Scale for object modeling in Scan AR

Figure 4.32 Scaling the object modeling size



When a user is viewing the object modeling, the user can click "scale up", “scale down” to view object modeling size increase or decrease display. Stop scale is when the user only wants the object modeling to stop scale when scaling the object. **Figure 4.33** shows the segment code for scaling the object.

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class AuhmentedReality_UI : MonoBehaviour
{
    public float scalingspeed = 0.09f;
    bool ScaleUp = false;
    bool ScaleDown = false;

    void Start()
    {

    }

    // Update is called once per frame
    void Update()
    {
        if (ScaleUp == true)
            ScaleUpButton();

        if (ScaleDown == true)
            ScaleDownButton();
    }
```

Figure 4.33 Segments of code for scaling objects

```
25
26      public void ScaleUpButton()
27      {
28          GameObject.FindWithTag("marker").transform.localScale += new Vector3(scalingspeed, scalingspeed, scalingspeed);
29          GameObject.FindWithTag("Model1").transform.localScale += new Vector3(scalingspeed,scalingspeed,scalingspeed);
30          GameObject.FindWithTag("ice").transform.localScale += new Vector3(scalingspeed, scalingspeed, scalingspeed);
31          GameObject.FindWithTag("marker1").transform.localScale += new Vector3(scalingspeed, scalingspeed, scalingspeed);
32          GameObject.FindWithTag("marker2").transform.localScale += new Vector3(scalingspeed, scalingspeed, scalingspeed);
33          GameObject.FindWithTag("markerextra").transform.localScale += new Vector3(scalingspeed, scalingspeed, scalingspeed);
34          GameObject.FindWithTag("marker4").transform.localScale += new Vector3(scalingspeed, scalingspeed, scalingspeed);
35          GameObject.FindWithTag("marker5").transform.localScale += new Vector3(scalingspeed, scalingspeed, scalingspeed);
36          GameObject.FindWithTag("marker6").transform.localScale += new Vector3(scalingspeed, scalingspeed, scalingspeed);
37      }
38
39      public void ScaleDownButton()
40      {
41          GameObject.FindWithTag("marker").transform.localScale += new Vector3(-scalingspeed, -scalingspeed, -scalingspeed);
42          GameObject.FindWithTag("Model1").transform.localScale += new Vector3(-scalingspeed, -scalingspeed, -scalingspeed);
43          GameObject.FindWithTag("ice").transform.localScale += new Vector3(-scalingspeed, -scalingspeed, -scalingspeed);
44          GameObject.FindWithTag("marker1").transform.localScale += new Vector3(-scalingspeed, -scalingspeed, -scalingspeed);
45          GameObject.FindWithTag("marker2").transform.localScale += new Vector3(-scalingspeed, -scalingspeed, -scalingspeed);
46          GameObject.FindWithTag("markerextra").transform.localScale += new Vector3(-scalingspeed, -scalingspeed, -scalingspeed);
47          GameObject.FindWithTag("marker4").transform.localScale += new Vector3(-scalingspeed, -scalingspeed, -scalingspeed);
48          GameObject.FindWithTag("marker5").transform.localScale += new Vector3(-scalingspeed, -scalingspeed, -scalingspeed);
49          GameObject.FindWithTag("marker6").transform.localScale += new Vector3(-scalingspeed, -scalingspeed, -scalingspeed);
50
51
```

Figure 4.34 Segments of code for each marker to set scale

```
56
57          public void Up()
58          {
59              ScaleUp = true;
60              ScaleDown = false;
61          }
62
63          public void Down()
64          {
65              ScaleUp = false;
66              ScaleDown = true;
67          }
68
69          public void Stop()
70          {
71              ScaleUp = false;
72              ScaleDown = false;
73          }
74      }
75
```

Figure 4.35 Segments of codes for scaling

4.7.5 Creating quiz questions

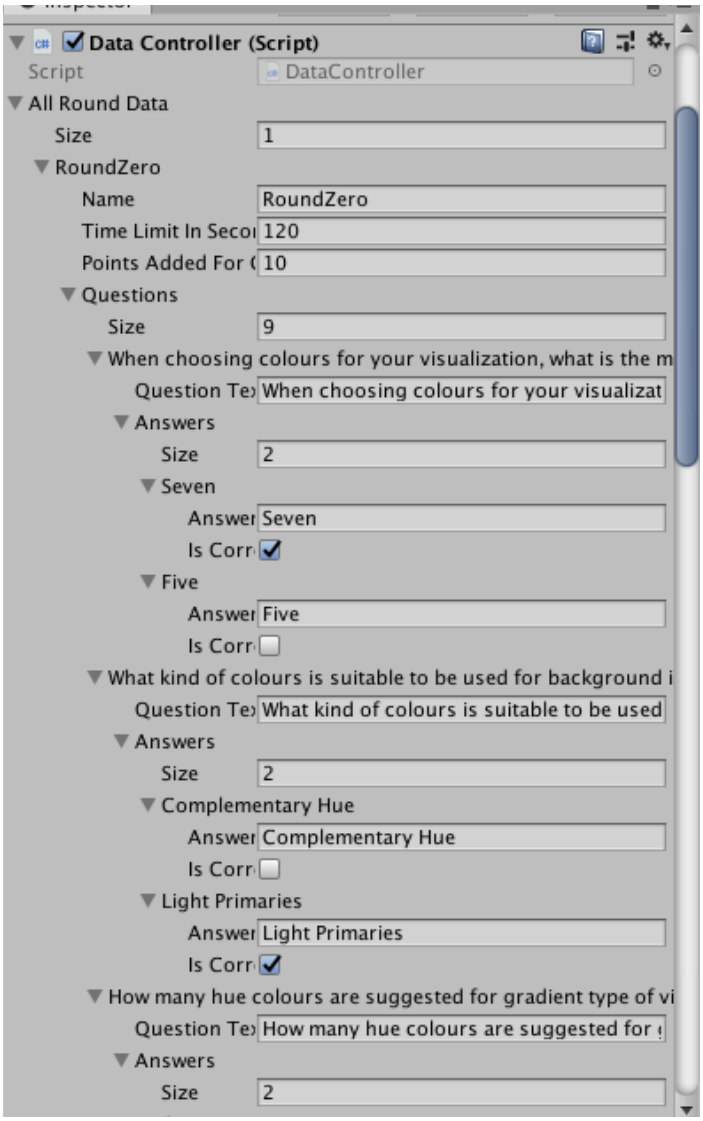


Figure 4.36 shows how questions are made in Unity and scripted

DataController.cs

selection

```
using UnityEngine;
using System.Collections;
using UnityEngine.SceneManagement;
using System.IO;

public class DataController : MonoBehaviour
{
    public RoundData[] allRoundData;
    private PlayerProgress playerProgress;

    // Use this for initialization
    void Start()
    {
        DontDestroyOnLoad(gameObject);
        LoadPlayerProgress();

        SceneManager.LoadScene("MainMenu");
    }

    public RoundData GetCurrentRoundData()
    {
        return allRoundData[0];
    }

    public void SubmitNewPlayerScore(int newScore)
    {
        if (newScore > playerProgress.highestScore)
        {
            playerProgress.highestScore = newScore;
            SavePlayerProgress();
        }
    }

    public int GetHighestPlayerScore()
    {
        return playerProgress.highestScore;
    }
```

```
        {
            playerProgress.highestScore = PlayerPrefs.GetInt("highestScore");
        }

    }
    private void SavePlayerProgress()
    {
        PlayerPrefs.SetInt("highestScore", playerProgress.highestScore);
    }


}
```

Figure 4.37 Segment of codes for quiz questions and quiz flow

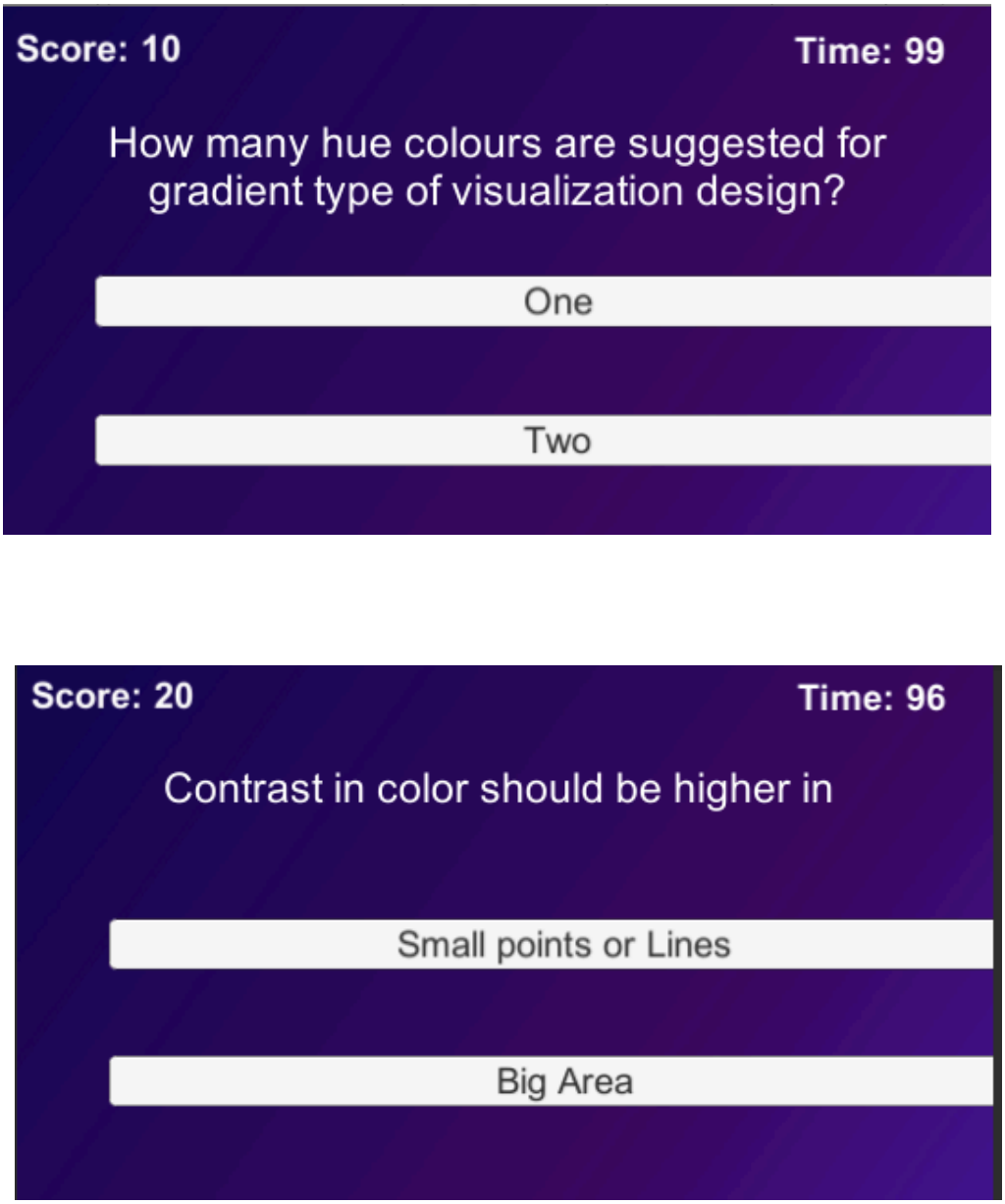


Figure 4.38 Output for example of questions display in quiz

4.7.6 Creating score and the highest score

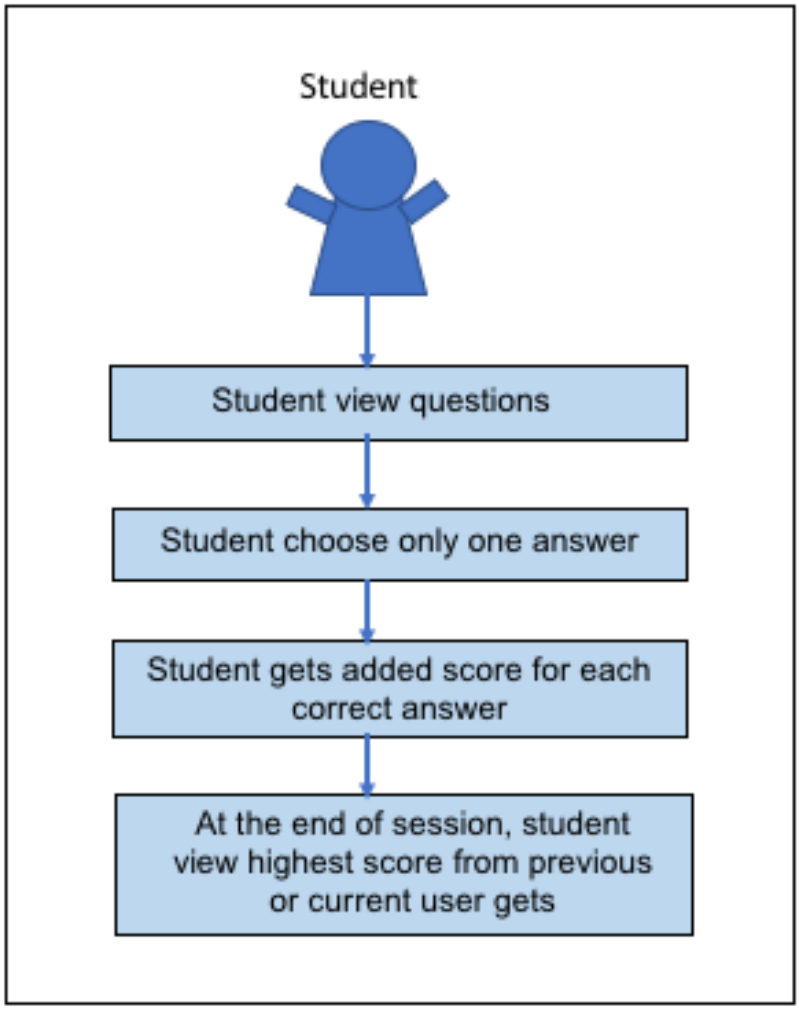


Figure 4.39 how score and highest score display

User will receive score when they answer each question correctly and view the highest score at the end of quiz session, Figure 4.40 shows the segment code of score and the highest score gets from the correct answer

selection

```
using UnityEngine;
using System.Collections;
using UnityEngine.SceneManagement;
using System.IO;

public class DataController : MonoBehaviour
{
    public RoundData[] allRoundData;
    private PlayerProgress playerProgress;

    // Use this for initialization
    void Start()
    {
        DontDestroyOnLoad(gameObject);
        LoadPlayerProgress();

        SceneManager.LoadScene("MainMenu");
    }

    public RoundData GetCurrentRoundData()
    {
        return allRoundData[0];
    }


    public void SubmitNewPlayerScore(int newScore)
    {
        if (newScore > playerProgress.highestScore)
        {
            playerProgress.highestScore = newScore;
            SavePlayerProgress();
        }
    }

    public int GetHighestPlayerScore()
    {
        return playerProgress.highestScore;
    }

    private void LoadPlayerProgress()
    {
        playerProgress = new PlayerProgress();

        if (PlayerPrefs.HasKey("highestScore"))
        {
            playerProgress.highestScore = PlayerPrefs.GetInt("highestScore");
        }

    }
    private void SavePlayerProgress()
    {
        PlayerPrefs.SetInt("highestScore", playerProgress.highestScore);
    }

```

Figure 4.40 segment code of score and the highest score

4.7.7 Adding time when answering quiz

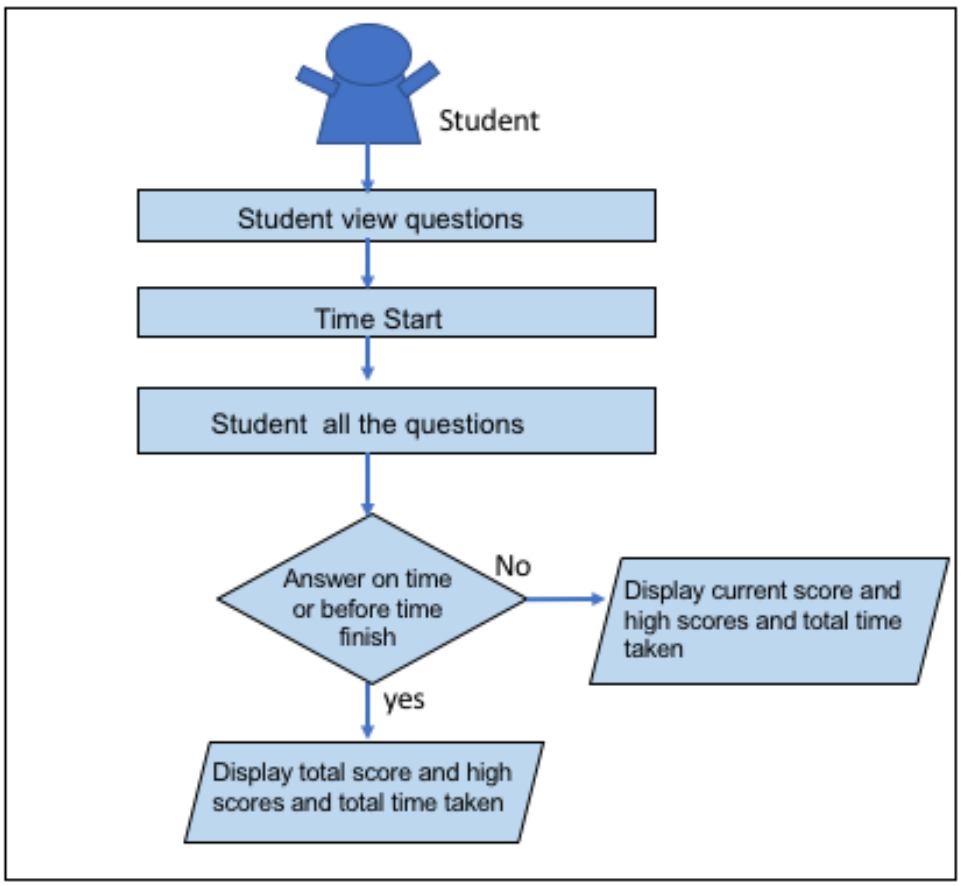


Figure 4.41 Time display

Users need to finish all the questions based on the time given in the quiz. User will display current mark if they did not finish answering questions on time and the total mark will display when the user finish on or before the time ends. Figure 4.42 shows segment code for time

```
    public void EndRound()
    {
        isRoundActive = false;

        dataController.SubmitNewPlayerScore(playerScore);
        highScoreDisplay.text = dataController.GetHighestPlayerScore().ToString();

        questionDisplay.SetActive(false);
        roundEndDisplay.SetActive(true);
    }

    public void ReturnToMenu()
    {
        SceneManager.LoadScene("MainMenu");
    }

    private void UpdateTimeRemainingDisplay()
    {
        timeRemainingDisplayText.text = "Time: " + Mathf.Round(timeRemaining).ToString();
    }

    // Update is called once per frame
    void Update()
    {
        if (isRoundActive)
        {
            timeRemaining -= Time.deltaTime;
            UpdateTimeRemainingDisplay();

            if (timeRemaining <= 0f)
            {
                EndRound();
            }

        }
    }
}
```

Figure 4.42 segment of code of time display and the remaining time

Figure 4.43 Output for score, high score, and time

4.7.8 Creating Videos procedure

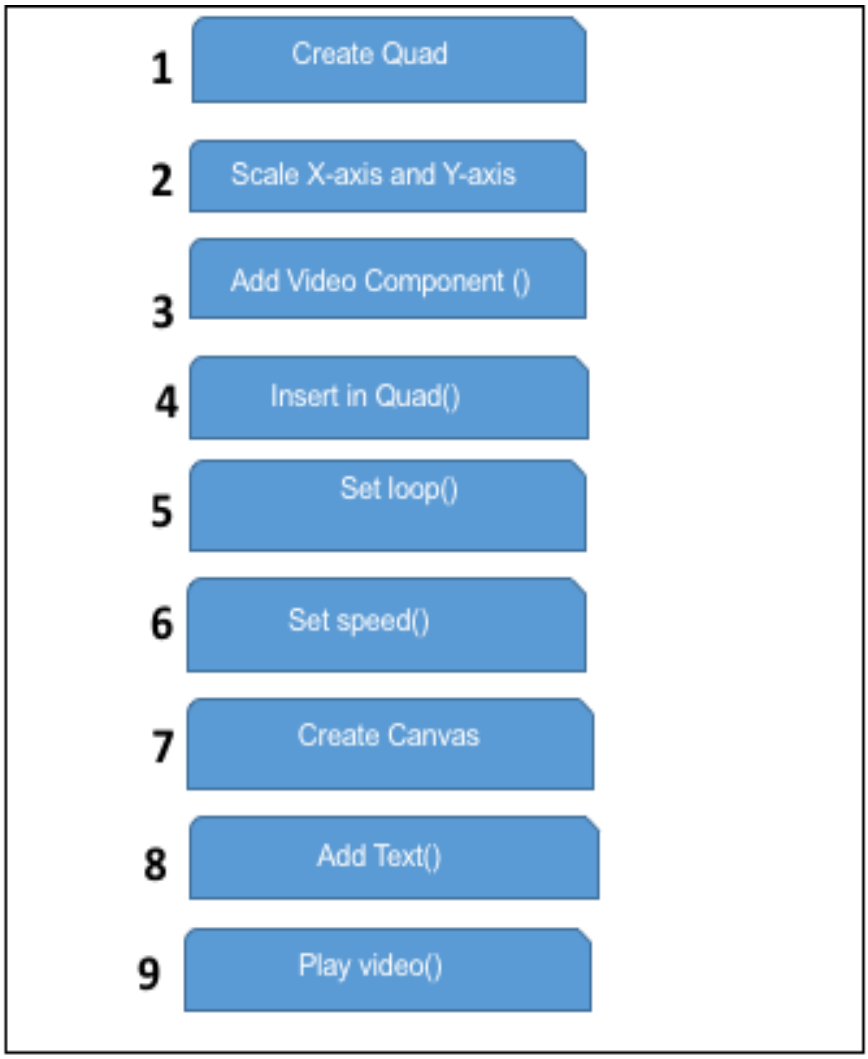


Figure 4.44 Creating Videos

There are nine steps in creating the video as shown in Figure above. The first step has begun with creating an empty quad. The quad works like a frame for only the videos to detect and display. The second step is to adjust the Scale X and Scale Y axis accordingly to make it easier for camera detection. The third step is to add a video component to be displayed. The video component is inserted in Quad that has been created in step fourth. For the video to be played repeatedly and in control speed, loop and speed need to set for able in step fifth and sixth. Based on steps seventh and eight, the canvas is created and the text was added for additional interaction. The last step is to test play the video to check if the video is correctly placed in the quad.

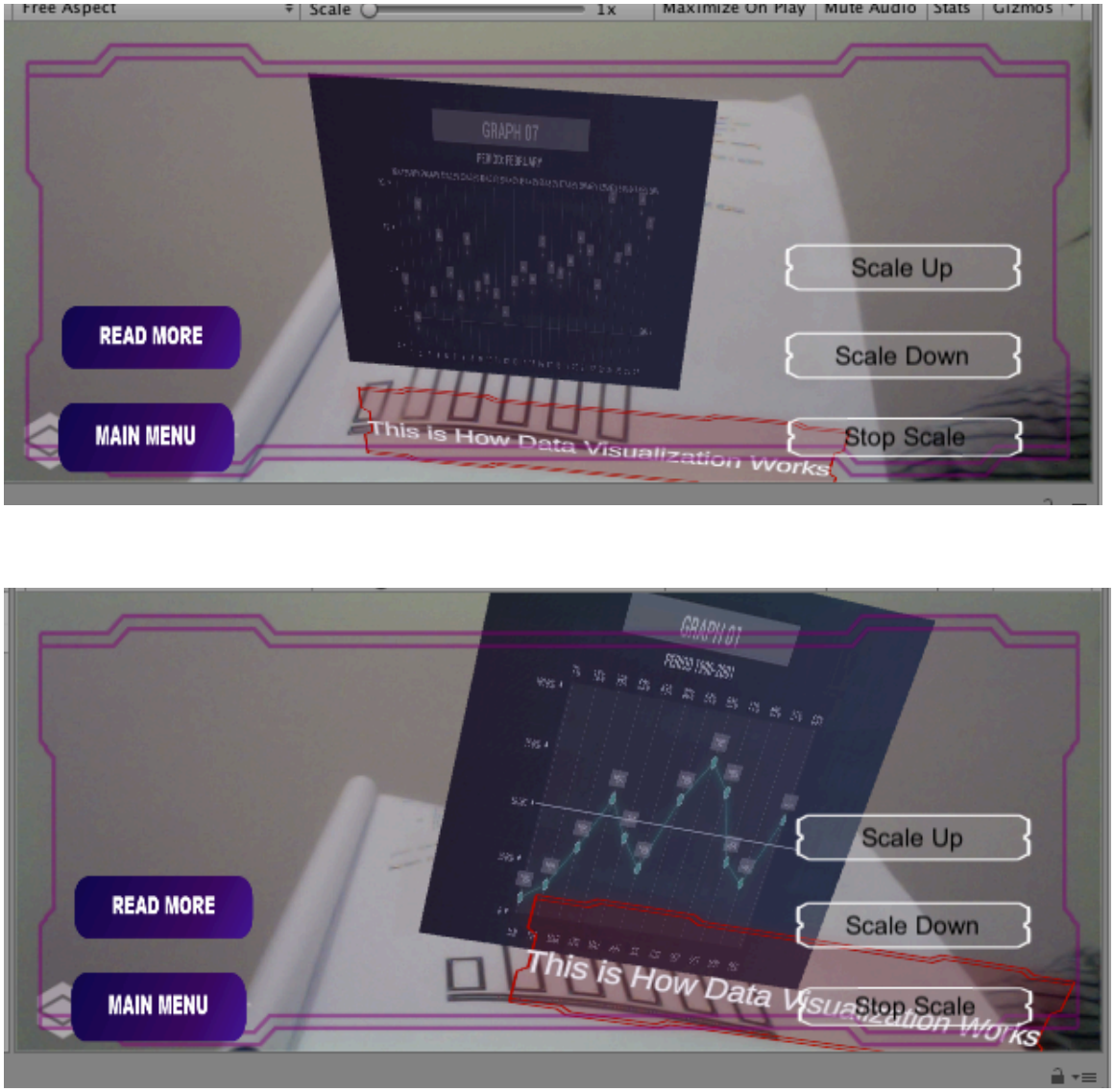


Figure 4.45 Output of Video play display

4.7.9 Creating Animation procedure

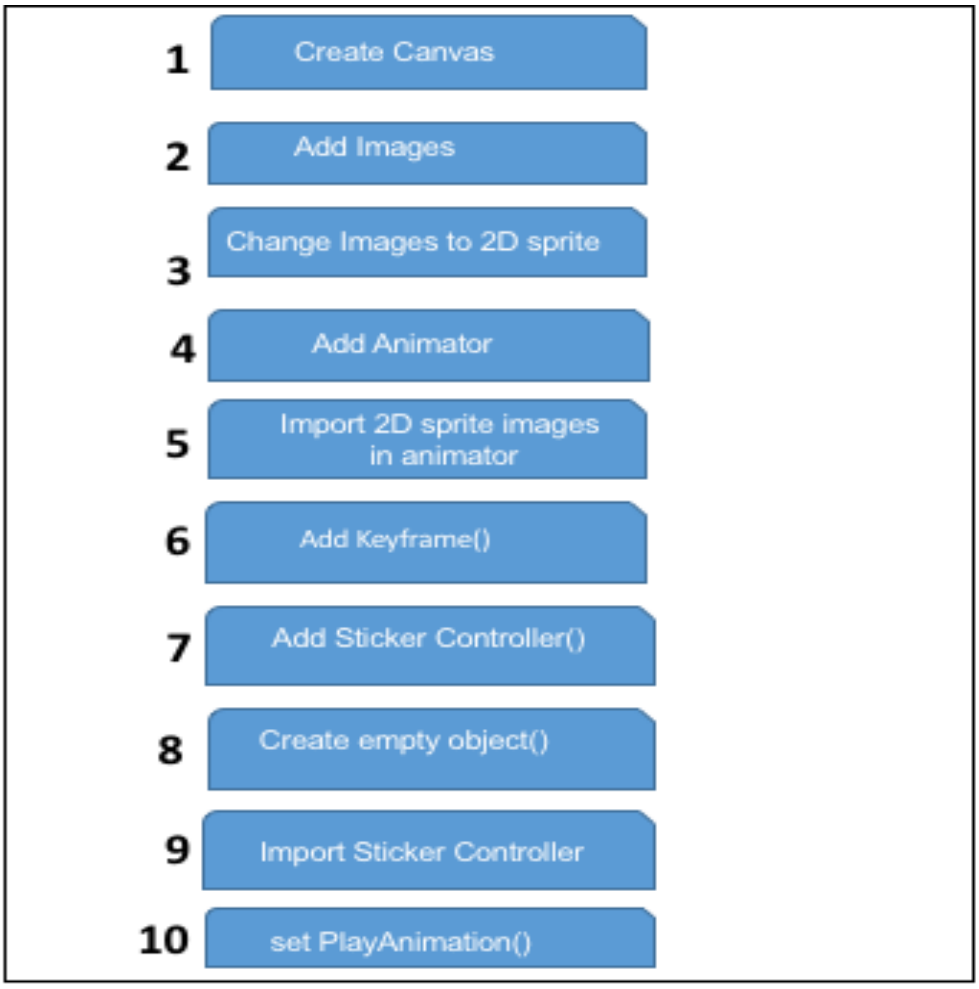


Figure 4.46 Creating Animation

There are ten steps in creating animation as shown in Figure above. The first step has begun with creating an empty canvas. The canvas works like a frame for the image to detect and display. The second step is to add images and change the images to be in 2D sprite as shown in step three. Following steps 4 and 5, animators are added and 2D sprite images are imported to be arranged in sequence. The keyframe is added in each image in the animator for the animation to be work. In the seventh step, a sticker controller is added where it works as a program to play the animation. An empty object is created in step eight and the sticker controller was import in step nine. The last step is to set play animation in sequence when the animation display. Figure 4.47 and Figure 4.48 show the process of how animation being

created while Figure 4.49 shows a segment of code of animation to play when objects being displayed.

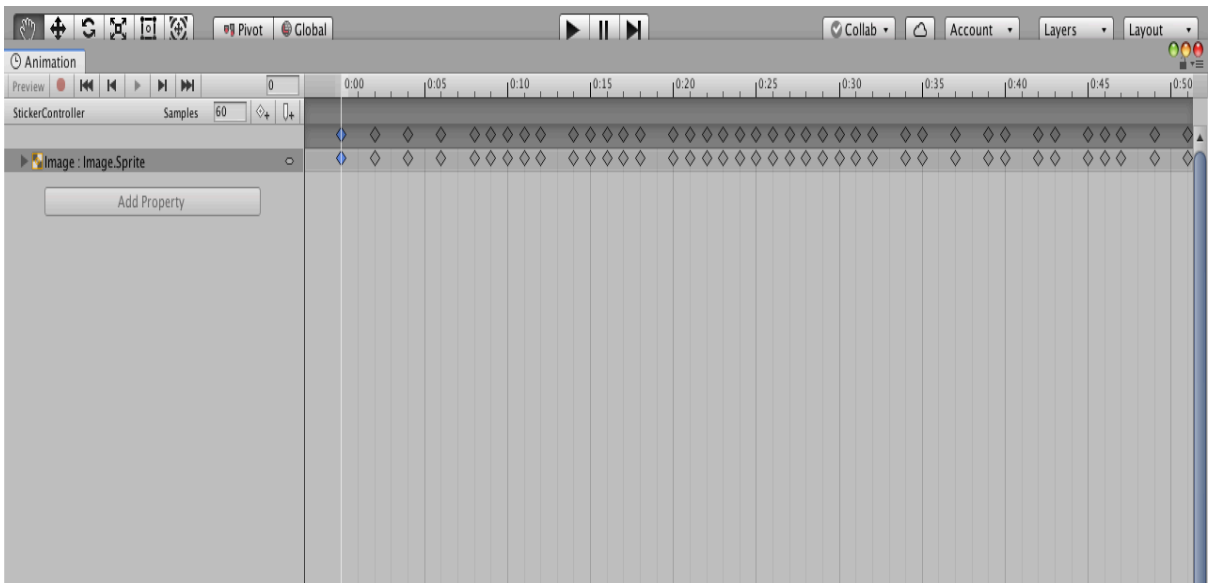


Figure 4.47 Keyframes for each image

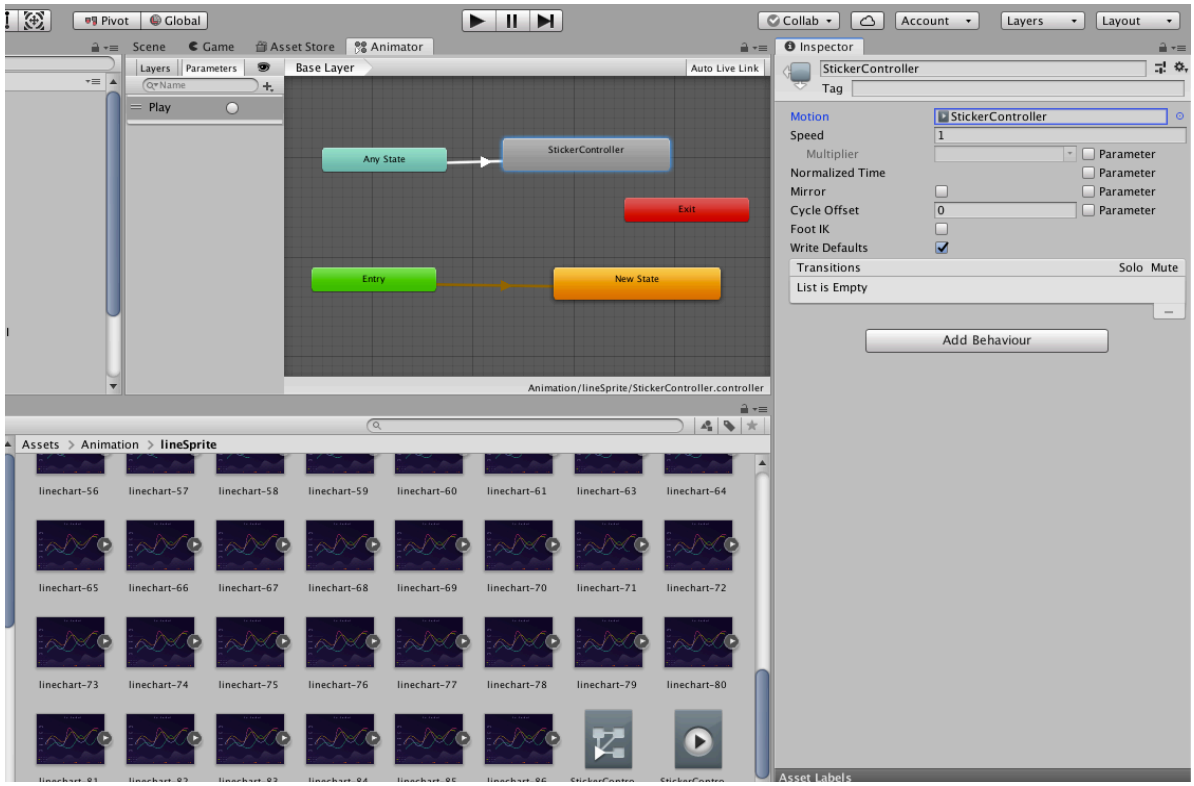
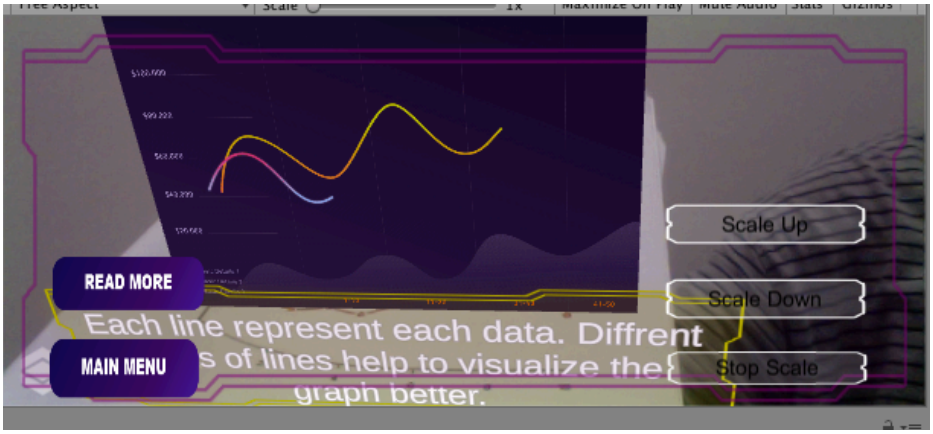
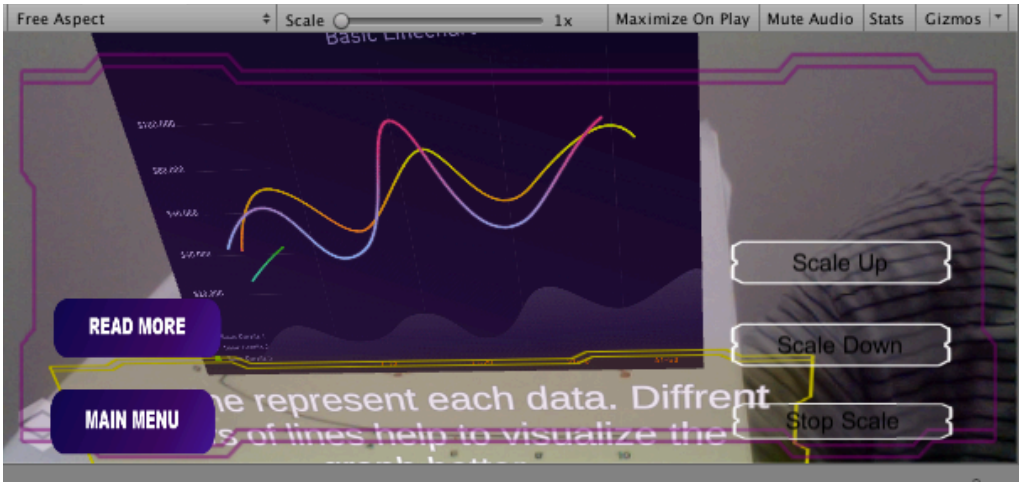


Figure 4.48 Sticker Controller for the animation to play in sequence

```
1   public class DefaultTrackableEventHandler : MonoBehaviour, ITrackableEventHandler
2   {
3
4       public Animator stickerAnimator;
5
6
7       #region PROTECTED_MEMBER_VARIABLES
8
9       protected TrackableBehaviour mTrackableBehaviour;
10      protected TrackableBehaviour.Status m_PreviousStatus;
11      protected TrackableBehaviour.Status m_NewStatus;
12
13   protected virtual void OnTrackingFound()
14      {
15
16          var rendererComponents = GetComponentsInChildren<Renderer>(true);
17          var colliderComponents = GetComponentsInChildren<Collider>(true);
18          var canvasComponents = GetComponentsInChildren<Canvas>(true);
19
20
21          // Enable rendering:
22          foreach (var component in rendererComponents)
23              component.enabled = true;
24
25          // Enable colliders:
26          foreach (var component in colliderComponents)
27              component.enabled = true;
28
29          // Enable canvas':
30          foreach (var component in canvasComponents)
31              component.enabled = true;
32
33          stickerAnimator.SetTrigger("Play");
34      }
```

Figure 4.49 Segments code of animation





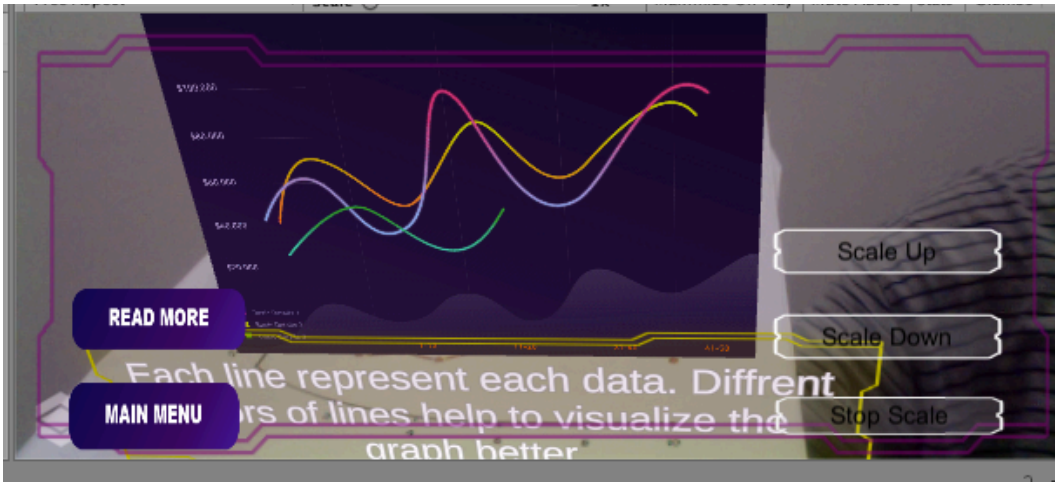


Figure 4.50 shows output of the animation

4.8 Conclusion

The design and development of this mobile application are based on the agile model. This mobile application system allows students to learn data visualization subjects in an interactive way using augmented reality by viewing the subject content through 3D modeling. To gain student's understanding, students need to answer the quiz. Score, highest score and time will be display for the user.

CHAPTER 5

TESTING/RESULT & FINDINGS

This chapter discuss about testing that has been conducted for this project and the testing result. Functionality testing is the type of testing that will be used to conduct in this project.

5.1 Overall System Functionality

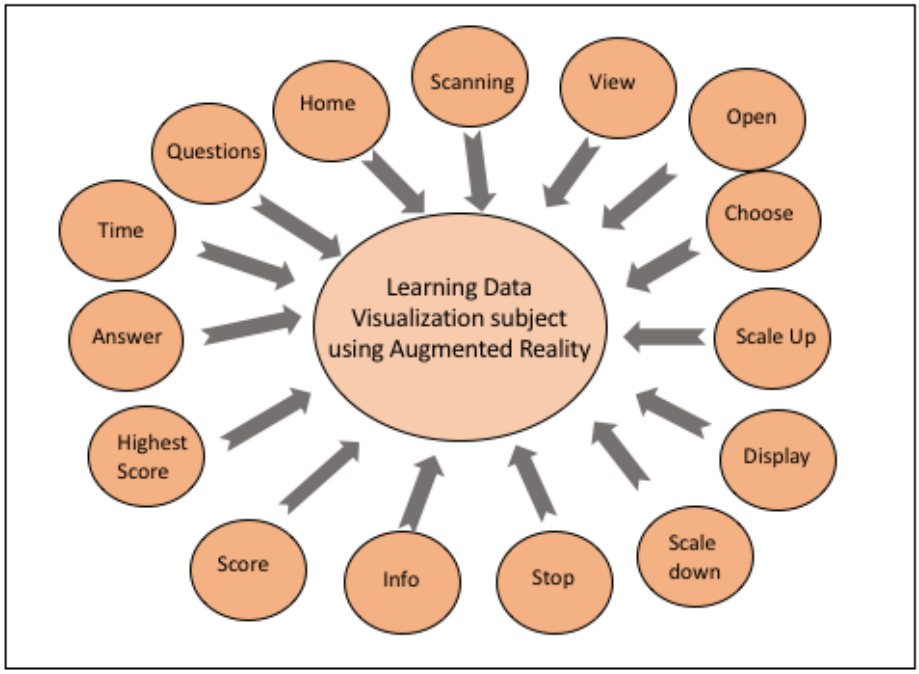


Figure 5.1 shows overall system functionality

Table 5.1 Description of overall main function

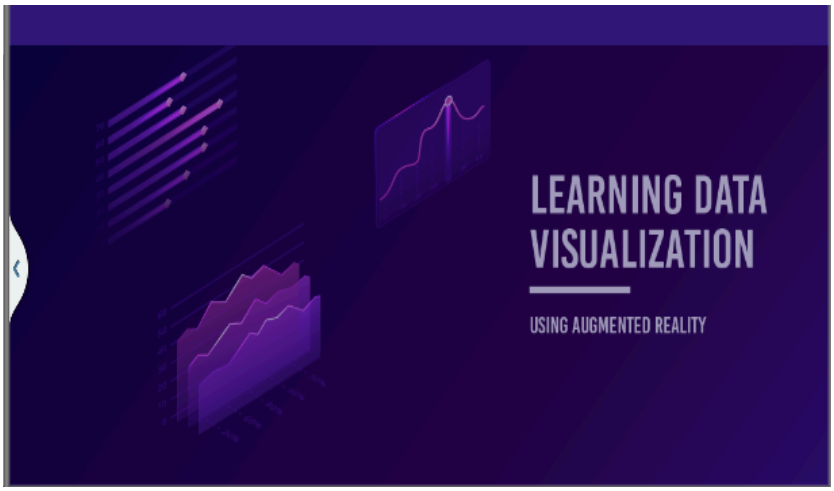
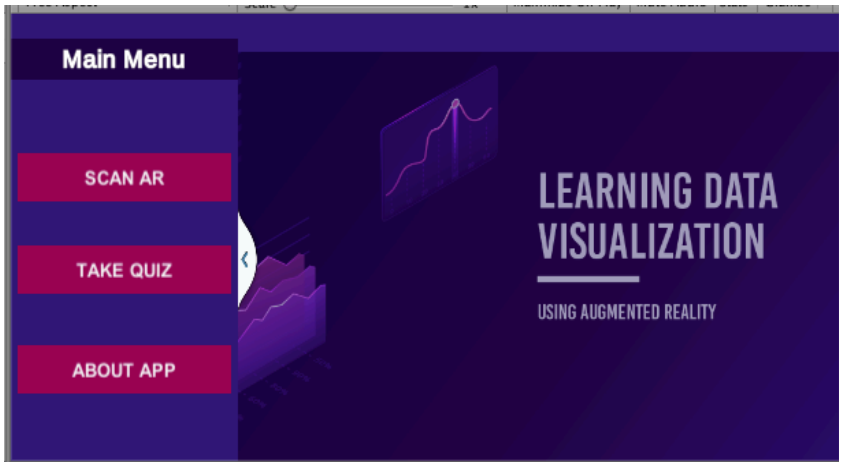
Function	**Description**
Open	In order to use this system, user must click toogle to open the the main menu
Camera	In order to view the modelling object, user must capture the object through camera function
Choose marker	Allow user to choose which marker they want to scan
Scanning marker	In order to display the object modelling, user need to scan marker for detection
Display object modelling	User will able to see object modelling display in the screen
Display video	User will be able to see object played in video
Display animation	User will be able to see object played in animation

View information	Allow user to view information about the modelling in details.
Scale Up object	Allow user to scale up the object modelling
Scale Down object	Allow user to scale down the object modelling
Stop Scaling	Allow user to stop scaling the object modelling.
Information about the application	This function is for first time user to know how to use the system and how the system works.
Quiz	Allow students to enhance their understanding by taking quiz
Score	This function is to display score that user gets after answering quiz
Highest Score	This function is to display highest score get from previous user
Answer	User need to choose one correct answer only

Time	This function display time for user to answer all the questions
Quiz Questions	This function display questions for students to answer related to what they learnt
Home	This function is to navigate user to main menu

5.1.1 User Interface for Toggle button

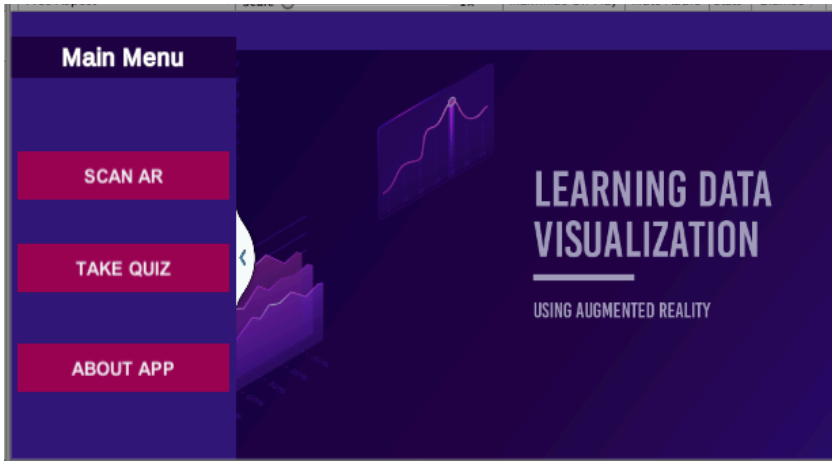
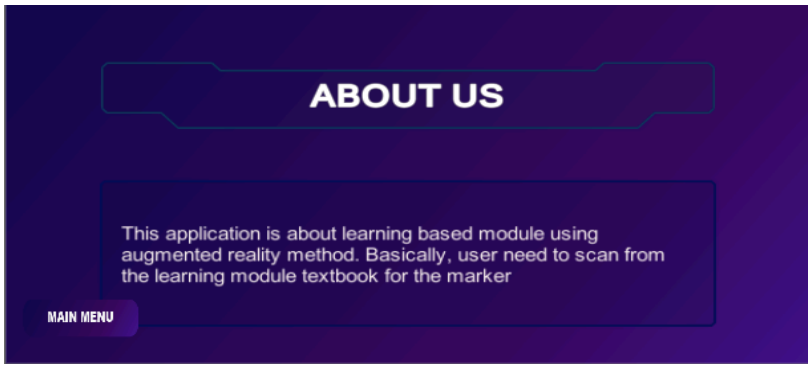
Table 5.2 Toggle button test

Test Objective	To test whether the function of toggle button can be performed
Potential Test Inputs	User click the toggle button 
Expected Test Outputs	
Test Procedures	User able to open the option for and click all buttons in the menu

Actual Test results	Toggle button are functioning in open and close the main menu

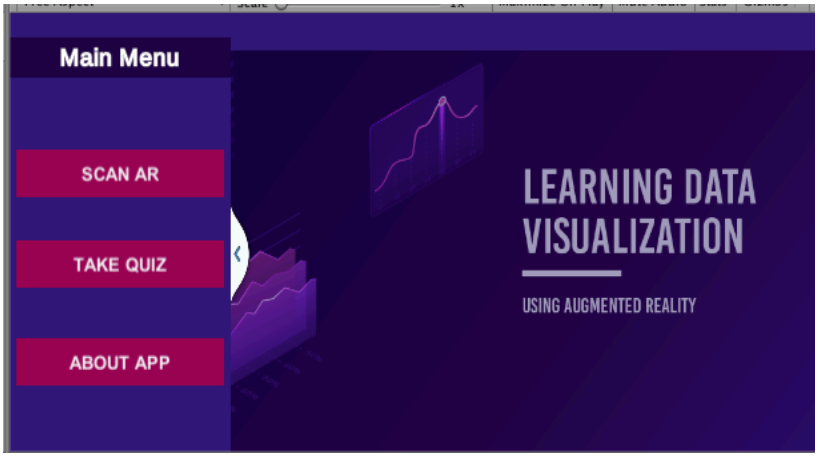
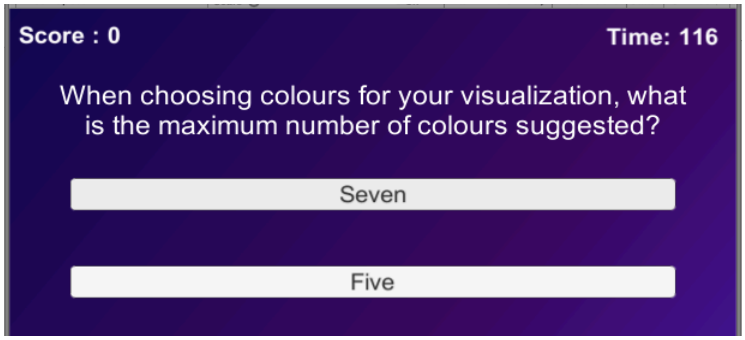
5.1.2 User Interface for About App

Table 5.3 View about application test

Test Objective	To test whether the function can be perform for user to be able read information and navigate to screen home button
Potential Test Inputs	Potential test input is able to view about application information 
Expected Test Outputs	
Test Procedures	View information about the application and able to navigate to main screen by click to “Main Menu”
Actual Test Results	User able to view the details information of the application.

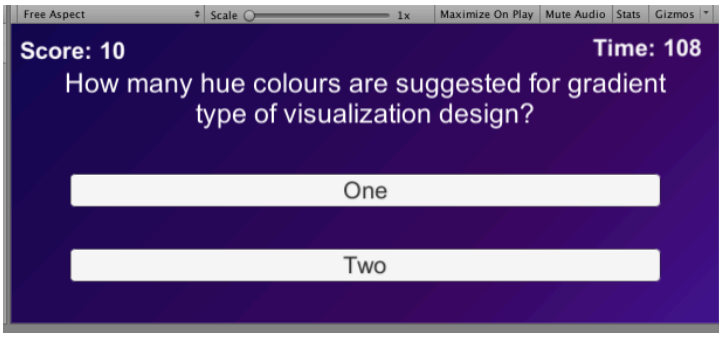
5.1.3 User Interface Take Quiz (Display Questions)

Table 5.4 Take Quiz

Test Objective	To test whether the function can be perform for user to be able to read and answer from the questions display
Potential Test Inputs	Potential test input is able for user to take quiz, read questions and choose answer. 
Expected Test Outputs	
Test Procedures	Questions is display to user and user need to choose
Actual Test Results	User need to choose only one correct answer from the questions display.

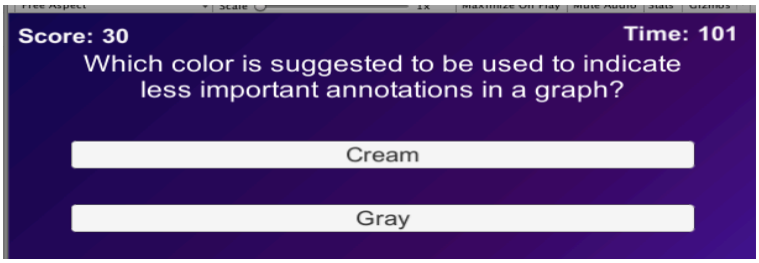
5.1.4 Time Display in quiz

Table 5.5 Time display in quiz

Test Objective	To test whether the function can be perform for user to be able to see time display and time remaining when they are answering the quiz.
Potential Test Inputs	Potential test input is time to start counting as soon as question is display
Expected Test Outputs	
Test Procedures	User need to answer the quiz within the time given
Actual Test Results	Time display and time remaining is display when user is answering the questions

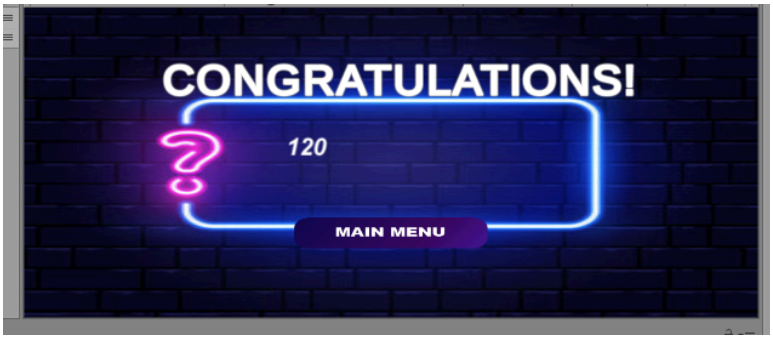
5.1.5 Score in quiz

Table 5.6 Display Score

Test Objective	To test whether the function can be perform for user to be able to see score added for the right questions or score remain for the wrong questions
Potential Test Inputs	Potential test input is score added or score remain when user choose the answer
Expected Test Outputs	
Test Procedures	User able to see their score when answering each questions in the quiz
Actual Test Results	Score added or Score remain after choosing one correct answer

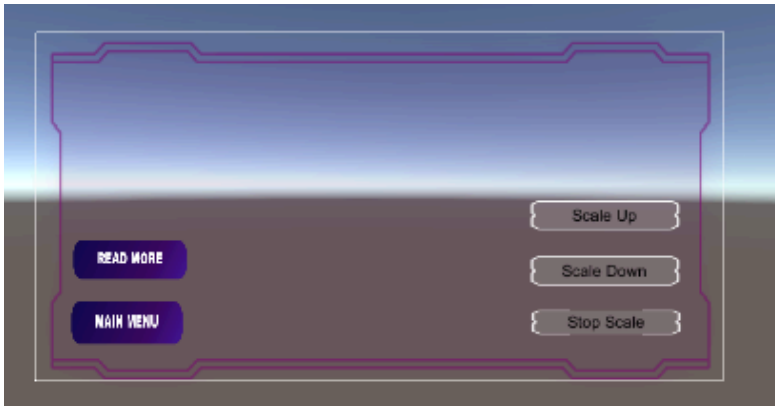
5.1.6 Highest score in quiz

Table 5.7 Display Highest Score

Test Objective	To test whether the function can be perform for user to be able to see highest score at the last session of answering all questions.
Potential Test Inputs	Potential test input is to display highest
Expected Test Outputs	
Test Procedures	User able to see their highest score at the end of quiz session
Actual Test Results	Highest score are able to be display

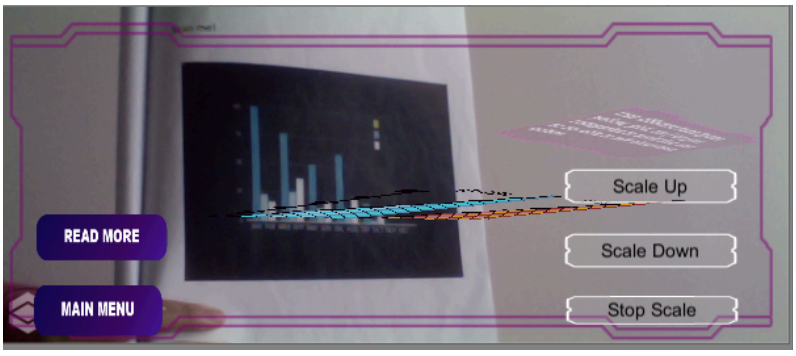
5.1.7 User Interface for Scan AR Camera

Table 5.8 Scan AR

Test Objective	• To test whether the function of camera functionality is working when user click the “Scan AR” button
Potential Test Inputs	The potential test input are camera to be display before doing the scanning 
Expected Test Outputs	
Test Procedures	User able to see and use the camera for the scanning.
Actual Test Results	Camera are display and able to functioning before doing the scanning

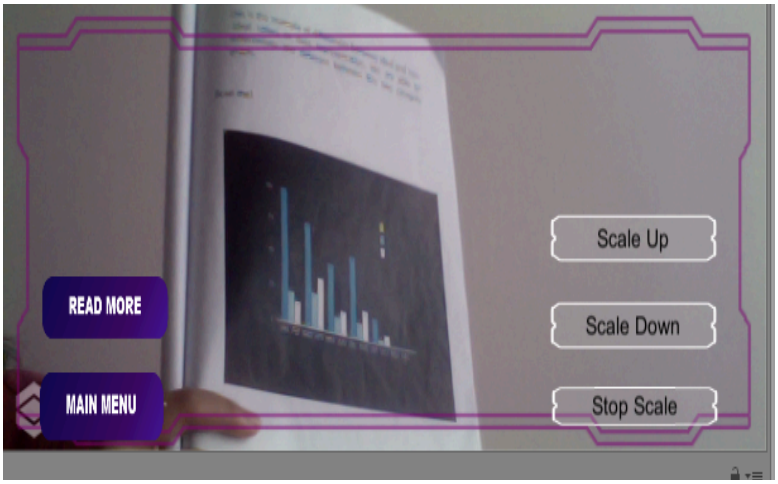
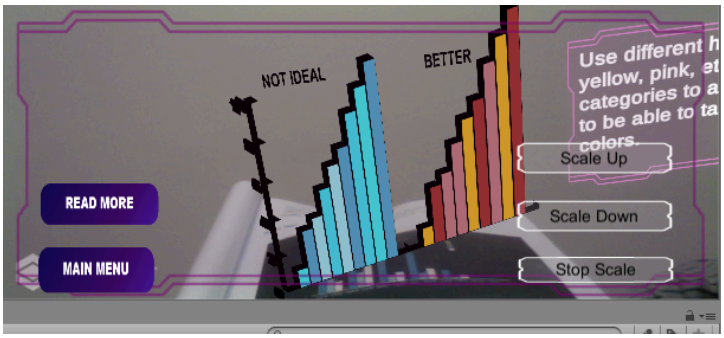
5.1.8 Scanning in Scan AR

Table 5.9 Scanning marker

Test Objective	• To test whether the camera are able to detect the marker through the scanning
Potential Test Inputs	The potential test input are marker detection through scanning by using the camera
Expected Test Outputs	
Test Procedures	User need to use the camera to scan the marker

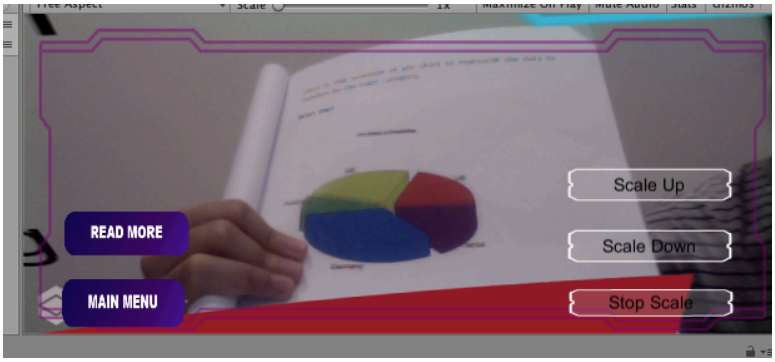
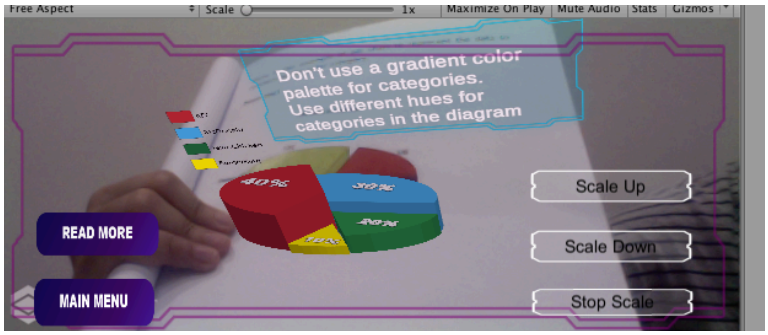
5.19 Display 3D object modelling

Table 5.10 object modelling

Test Objective	To test whether the 3D object modelling are display on screen after marker detection from the camera
Potential Test Inputs	Potential test input is 3D object modelling to be display 
Expected Test Outputs	
Test Procedures	User able to see 3D object modelling display in screen after scanning the marker
Actual Test Results	3D object modelling display as soon as camera detect the marker.

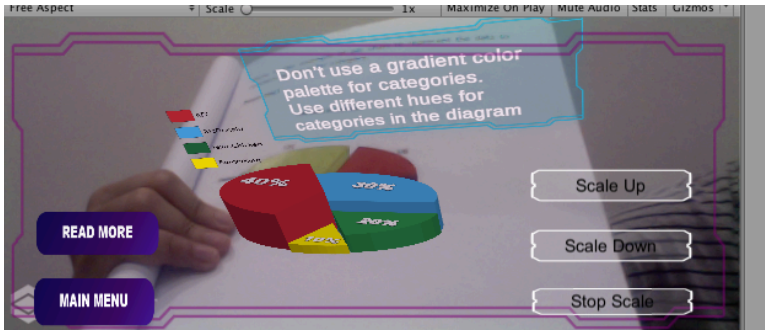
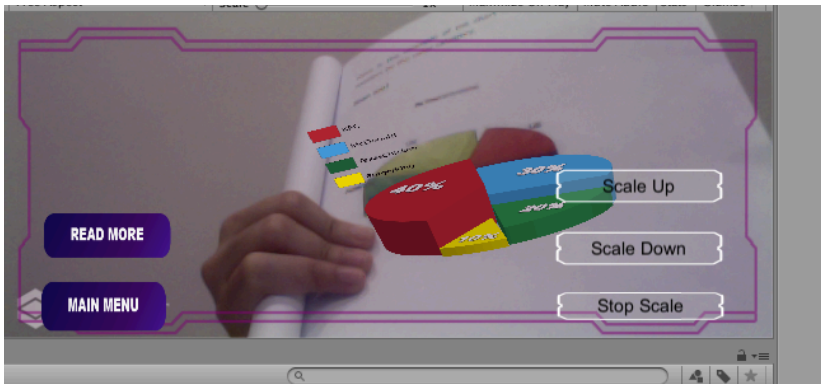
5.20 View info

Table 5.11 Display Info

Test Objective	To test whether the function can be perform for user to be able to read and view info from the 3D object modelling display
Potential Test Inputs	Potential test input is additional information display with the 3D object modelling 
Expected Test Outputs	
Test Procedures	User able to see 3D object modelling display with information display in screen
Actual Test Results	3D object modelling display with information as soon as camera detect the marker.

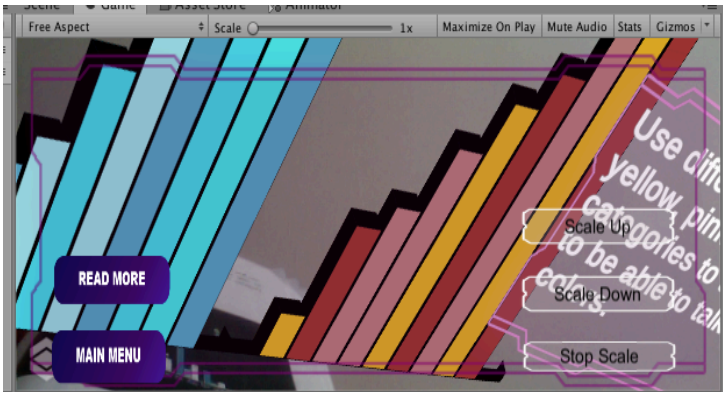
5.21 Hide info

Table 5.12 Hide Info

Test Objective	To test whether the function can be perform for user to be able to hide the information display and only view the object modelling
Potential Test Inputs	Potential test input is to hide information that display with the 3D object modelling 
Expected Test Outputs	
Test Procedures	User able to choose to hide the information that display with 3D object modelling or only view the 3D object modelling in screen
Actual Test Results	3D object modelling display with information or hide information depending on user preference

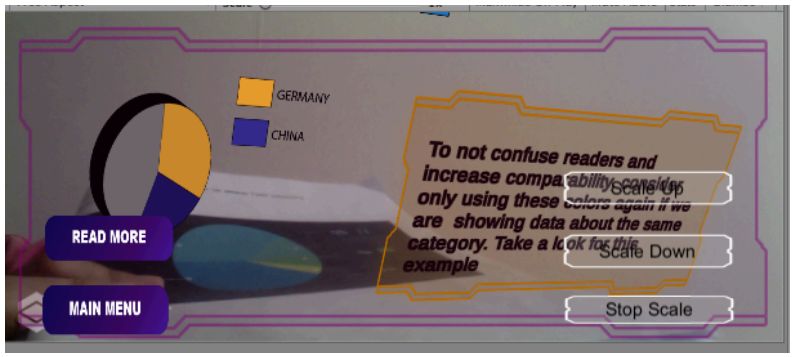
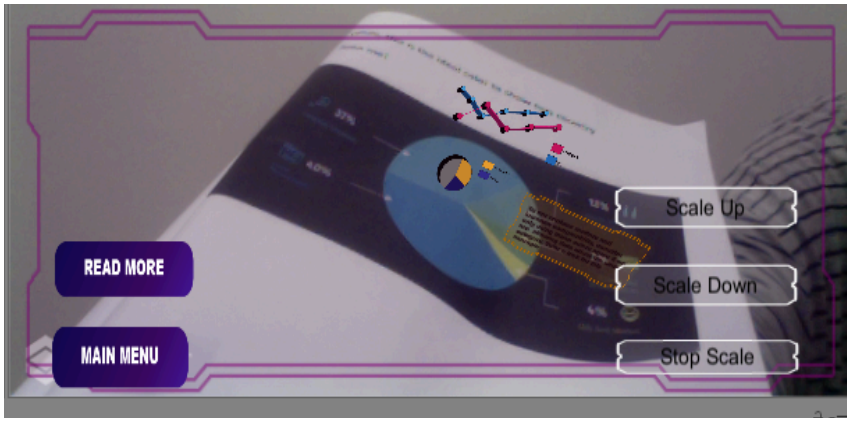
5.22 Scale Up

Table 5.13 Display scale up

Test Objective	To test whether the function can be perform for user to interact with the 3D modelling by scale up the size of 3D object that display on screen
Potential Test Inputs	Potential test input is the size of 3D modelling object increase
Expected Test Outputs	 
Test Procedures	User able to interact with the 3D object modelling by increasing the size of the object on screen
Actual Test Results	3D object modelling display with a larger size

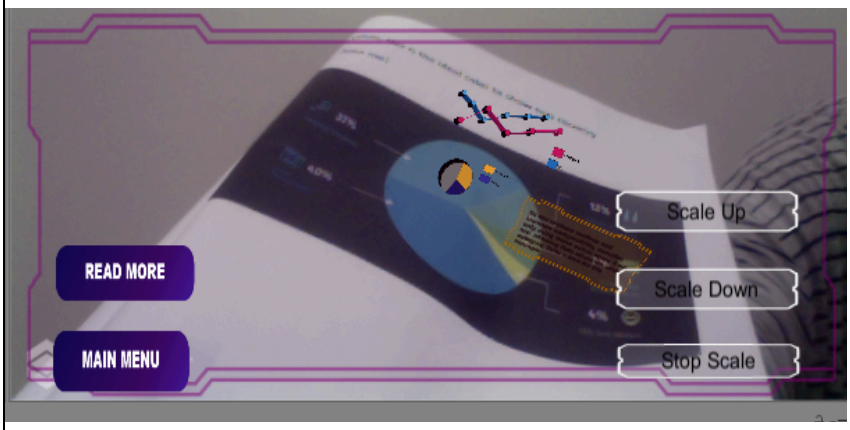
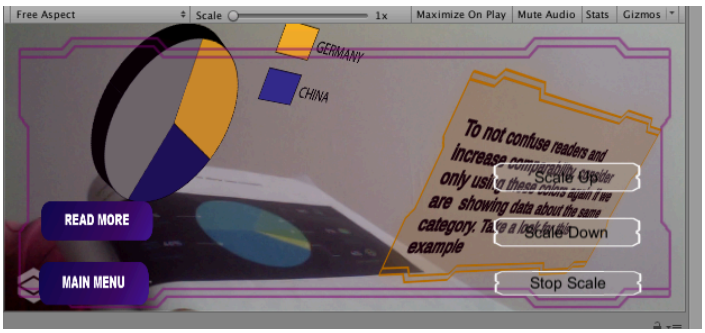
5.23 Scale Down

Table 5.14 display scale down

Test Objective	To test whether the 3D object modelling are display after marker detection from the camera
Potential Test Inputs	Potential test input is 3D object modelling to be display 
Expected Test Outputs	
Test Procedures	User able to see 3D object modelling display in screen after scanning the marker
Actual Test Results	3D object modelling display as soon as camera detect the marker.

5.25 Scale Stop

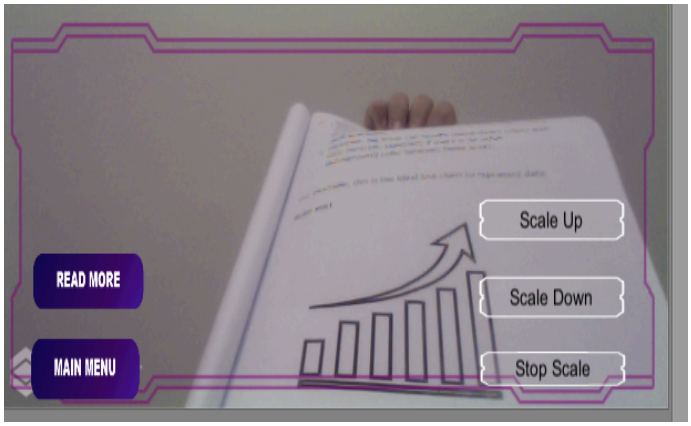
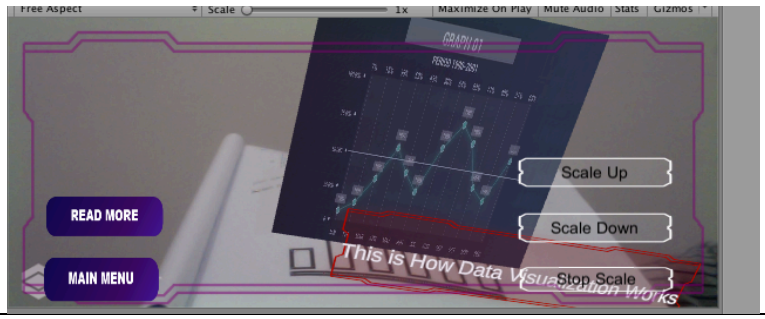
Table 5.15 display scale stop

Test Objective	To test whether the 3D object modelling are able to stop when scalling up or scalling down
Potential Test Inputs	Potential test input is 3D object modelling to be stop scaling 
Expected Test Outputs	
Test Procedures	User able to see 3D object modelling display stop scaling on screen

Actual Test Results	3D object modelling display stop scale up or scale down when user click “Stop Scale”

5.26 Display Video

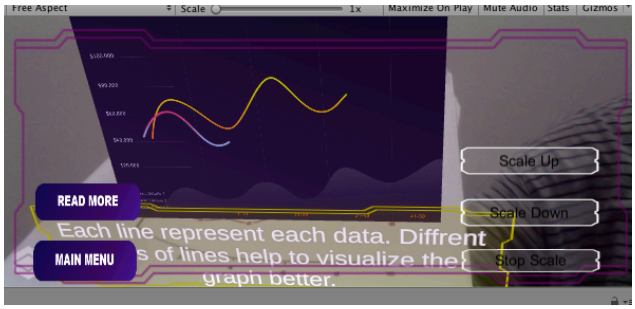
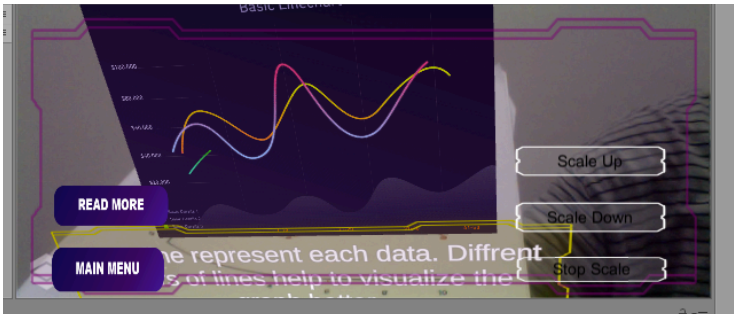
Table 5.16 display video

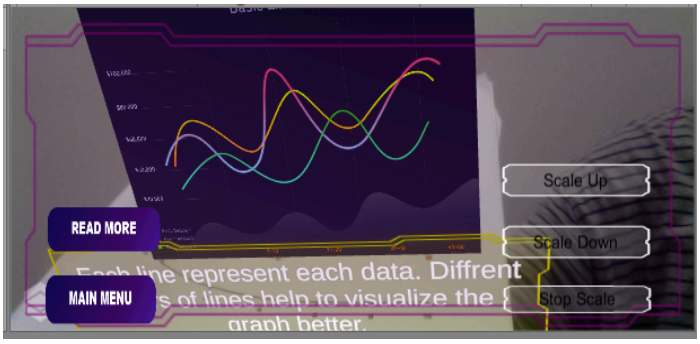
Test Objective	To test whether the 3D object modelling are able to play in video on screen
Potential Test Inputs	Potential test input is 3D object modelling to be play in video format 
Expected Test Outputs	

Test Procedures	User able to see 3D object modelling display stop scaling on screen
Actual Test Results	3D object modelling display stop scale up or scale down when user click “Stop Scale”

5.27 Play animation

Table 5.17 display animation

Test Objective	To test whether the 3D object modelling are able to animation on screen
Potential Test Inputs	Potential test input is 3D object modelling to be play in animation 
Expected Test Outputs	

	
Test Procedures	User able to see 3D object modelling display stop scaling on screen
Actual Test Results	3D object modelling display stop scale up or scale down when user click “Stop Scale”

5.2 Conclusion

This mobile application system has been tested by using functionality test. Every single unit function in this mobile application system has been tested as whole in order to determine and to fix the system error. The purpose of functionality testing is to ensure that all the development that has been made run as expected, all buttons are all functioning well and run without error.

CHAPTER 6

CONCLUSION AND FUTURE RESEARCH

This chapter discuss about the conclusion, limitation and future research of this project. All phases of developing this project are summarized. First section of this chapter discuss about summarization of this project as whole, followed by the limitation of this project and the third section discuss about the future enhancement and recommendation of this project.

6.1 Conclusion

As conclusion, this project is about to develop a learning mobile application for data visualization subject using extra features of Augmented Reality which called as Learning Data Visualization Subject using Augmented Reality. The purpose of this mobile application system is to allow students to learn in interactive way and enhance their interested towards this subject. Also, the development of this mobile application helps student to learn in their own path rather than just studied in static way. Quiz session were added in order to gain more student understanding and the outcome. All of the design and development of this project was developed and the stated objective in this project was achieved based on the objective that has been stated in chapter one. Objective of this project are very important in order to meet the requirement for this project.

In chapter two, research are made to develop this project and to gives insight on how to work with the project. In chapter three, the chosen methodology which is Agile model is as guidelines for the design and development of this project. Methodology can give better understanding about objectives and scope of this project and generally emphasized the flow of the process involves in the project.

In chapter four, the overall design and development phase of this project was developed based on the overall system functionality. In this phase, all the main function of this project are well defined.

While, in chapter five is about the testing that has been conducted. This Project tested by using functionality test. The function of this mobile application system are tested as whole, to ensure that all the requirement and needs from user are achieved and works as expected. In this chapter six, discuss about the limitation and future enhancement, so that this project can be enhanced and improved in the future as well as to overcome the limitation and weaknesses of this web based system. In order to develop this project, this project are using C# code to design the interface and also for the functions for each features.

A lot of research and practice are needed in doing this project especially gaining a skills in mobile application development. Overall, guidance from experts and supervisor are needed for us to have more knowledge in developing a system.

6.2 Limitation

This learning mobile application may not good as a perfect system but, all of the function that have been developed in this mobile application system are working as expected and the objective of this project are achieved. However, there are several weaknesses and limitation of this web based system such as follow:

- Difficult for students to takes notes when viewing the output
- Lecturers are still needed in order to explain for the topic more details
- Camera are only available in portrait mode
- Students cannot start the learning without the learning module to do the scanning

6.3 Future Recommendation

As a future enhancement or recommendation, more function are need to be added into this web based system in order to overcome the stated limitation. Some of the future enhancement that can be added are as follow:

- Retrieving the feedback for the quiz questions that student answers through emails
- Students able to screen capture the output content so that they can keep the picture for revision
- Adding a database to make sure who participate in learning the topic module
- Notification

  Adding notification function, so that student are alert when the time for answering the quizzes care about to finish

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